# Incentivizing Retail Traders: Evidence from Daily High-Water Marks on a Social Trading Platform

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# Technology is Changing How Retail Investors Trade

- Fintech has changed how consumers interact with financial markets
  - Spending (D'Acunto et al. 2019b) and saving (Gargano and Rossi 2024)
  - Robo-advisors help investors diversify (D'Acunto et al. 2019a)
- Platform design shapes the decisions retail investors make
  - 'Gamified' interfaces alter investor attention (Barber et al. 2022; Tan 2021)
- *A growing and relatively unexplored innovation: social trading platforms*

# Social Trading: Retail Traders as Fund Managers

- Retail **traders** make their investment decisions public
  - Retail **investors** buy an exchange-traded product that tracks the trader's portfolio
  - Retail traders effectively become fund managers
- The platform pays traders a performance bonus
  - A fraction of the fund's return above its previous high close
  - Earned any day the portfolio closes above that high, and the threshold resets immediately
- *An institutional high water mark contract, but paid daily to retail traders* (Ma et al. 2019; Aragon and Nanda 2012)

# How Could Daily Incentives Affect Retail Traders?

- Alignment
  - Performance pay aligns trader and investor interests, potentially mitigating retail biases
    - (Jensen and Meckling 1976; Holmstrom 1979)
- Or amplification
  - Extreme frequency amplifies loss aversion, excessive trading, and short-term thinking
    - (Odean 1998, 1999; Dorn et al. 2015; Delfabbro et al. 2021)
- **This paper: How does a daily high water mark bonus change the behavior and performance of retail traders?**

# Data and Methodology

- Data
  - **Social trading platform:** every trade, daily fund prices, and bonus days from 2018 to 2022
  - **Sample:** 5,147 funds and 3.6 million fund-day observations
  - **CRSP and LSEG:** returns of the underlying assets
  - *Increasing our coverage with German exchange data*
- Methodology
  - Regression discontinuity around the bonus threshold
  - Differences-in-differences around each bonus (de Chaisemartin and D'Haultfoeuille 2026)
  - January 1st reset of the high water mark
  - Counterfactual static portfolio where traders do not trade at all

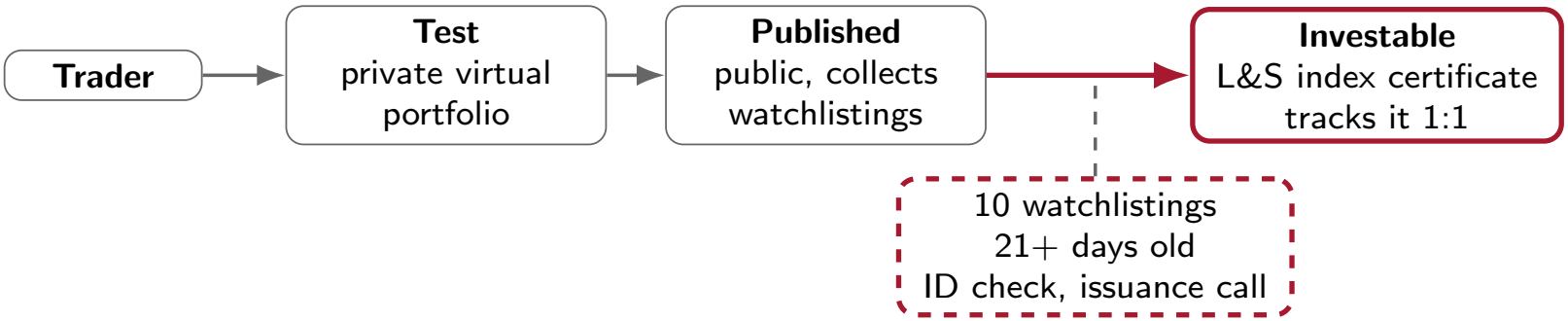
# Preview of the Main Results

- On days a bonus is achieved
  - Returns spike 1.6 percentage points above the previous day
    - Immediately fall back to pre-bonus levels
  - Odds of trading are 15% higher just above the bonus threshold
  - Traders sell recent winners to lock in the gain, but miss further run up
- Doing nothing would be better: 63% of funds would earn higher alpha and 57% of fund-years would earn more bonuses
- Consistent with daily incentives amplifying short-term focus rather than aligning interests

# Contribution

- Performance incentives are well studied for institutional managers (Brown et al. 1996; Massa and Patgiri 2009; Agarwal et al. 2009; Ray 2009); pay can align interests (Jensen and Meckling 1976; Holmstrom 1979)
  - Daily HWM amplifies short-term focus among retail traders
- Fintech platform design can help (D'Acunto et al. 2019a; Rossi and Utkus 2021; Gargano and Rossi 2024) or hurt users (Barber et al. 2022; Tan 2021)
  - Wall Street style incentives backfire at a daily frequency
- Retail and social trading (Barber and Odean 2000; Petajisto 2013; Anginer et al. 2024; Oehler et al. 2016; Roder and Walter 2019; Oehler and Schneider 2023)
  - Retail–professional gap persists even with performance pay

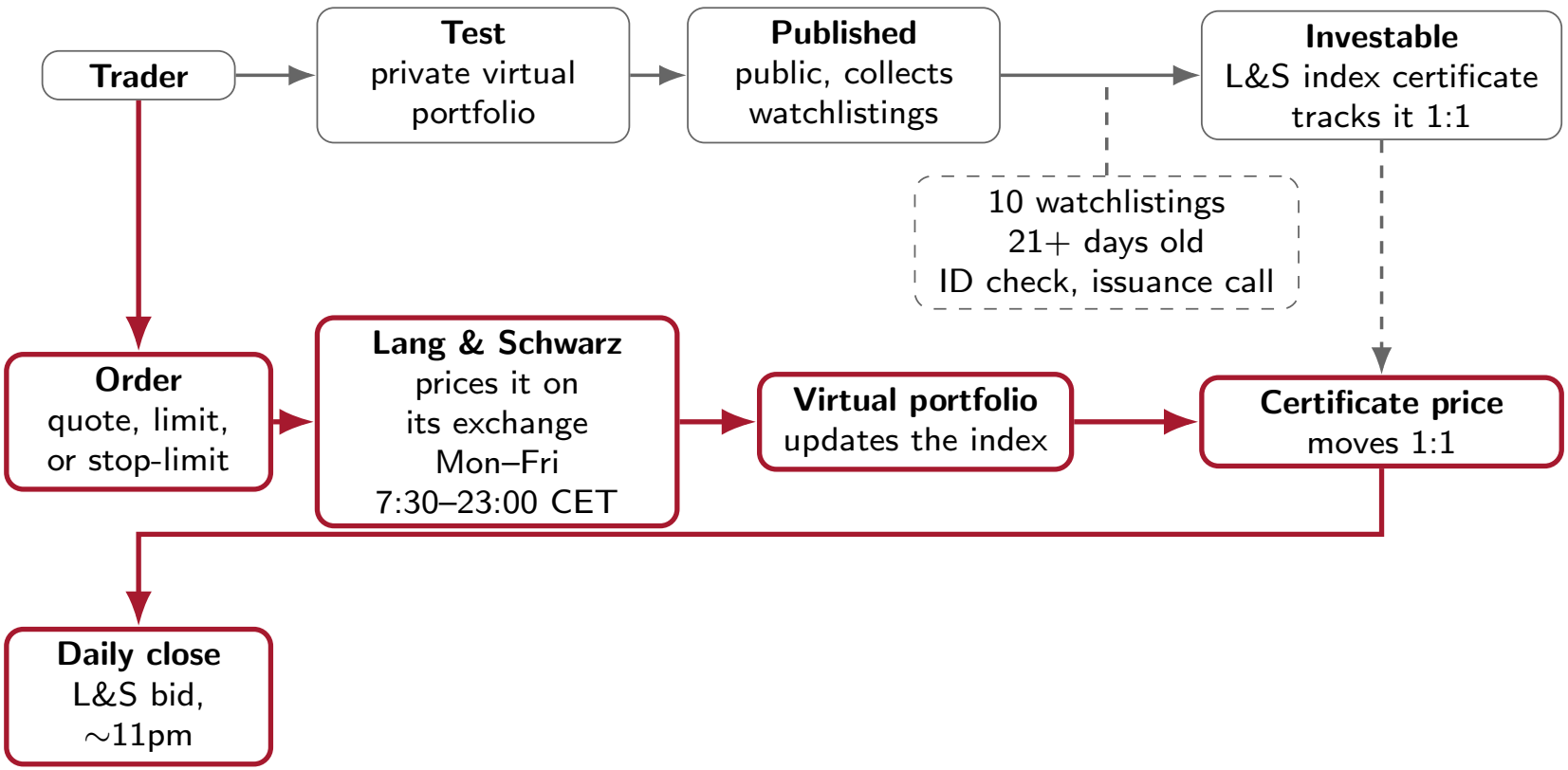
# How a Wikifolio Works



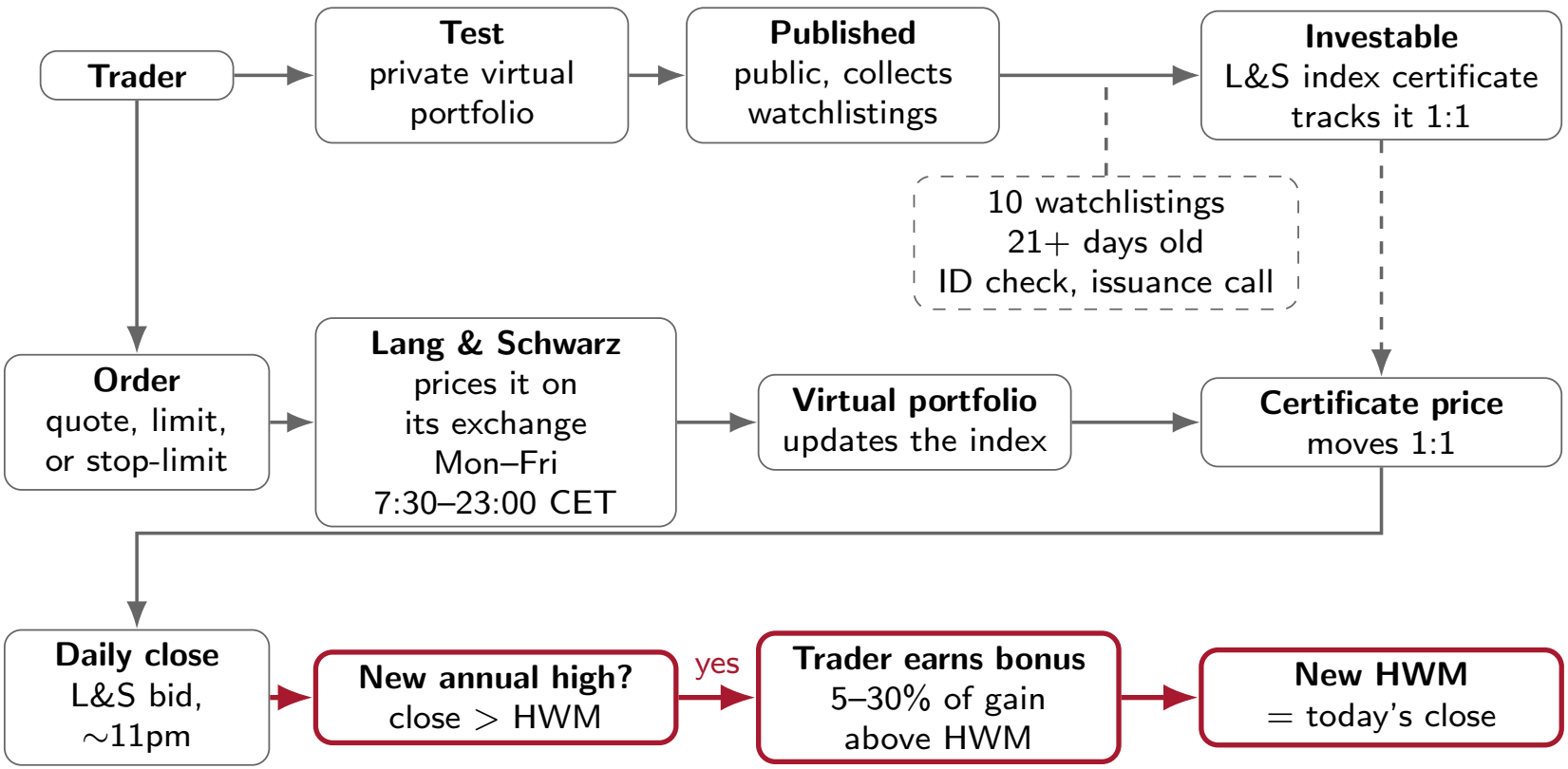
# Trading

- Trades are virtual, priced at real-time quotes for the underlying asset on the exchange run by Lang & Schwarz
  - Quote orders (price valid for 3 seconds), limit orders, and stop-limit orders
- Trading hours (CET): Mon–Fri 7:30am–11pm, Sat 10am–1pm, Sun 5–7pm
- No transaction costs for trades inside the wikifolio
- With invested capital, Lang & Schwarz hedges the positions
- Daily close = Lang & Schwarz bid price at about 11pm

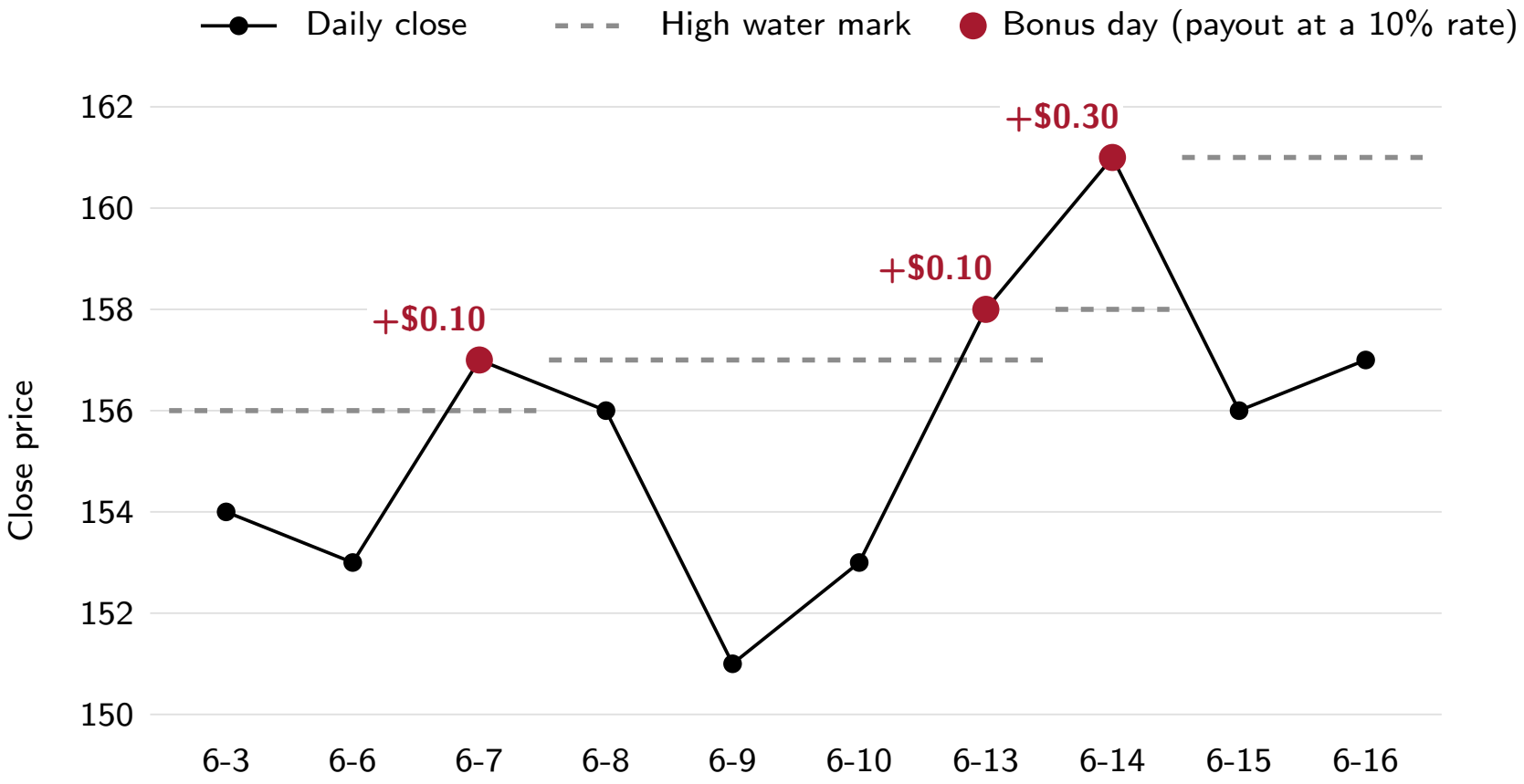
# How a Wikifolio Works



# How a Wikifolio Works



# Sample High Water Mark




Bonus = (Close − HWM) × Rate

# Portfolio Characteristics

| | Mean | Median | Standard Deviation |
|---|---|---|---|
| Daily Trades | 1.17 | 0 | 4.94 |
| Daily Buys | 0.66 | 0 | 3.02 |
| Daily Sales | 0.51 | 0 | 2.49 |
| Daily Trades when Trades $>$ 0 | 5.43 | 3 | 9.48 |
| Daily Quantity when Trades $>$ 0 | 5,206 | 100 | 349,310 |
| Daily Dollar Amount when Trades $>$ 0 | \$24,150 | \$2,086 | \$436,231 |
| Portfolio Life (Years) | 2.65 | 2.37 | 1.49 |
| Daily Returns | 0.02% | 0% | 2.05% |
| Daily Alpha | -0.0002% | -0.0001% | 0.37% |
| Performance Fee | 10.78% | 10% | 6.29% |
| Bonuses Earned Per Year | 30.1 | 20 | 32.6 |

About 1 trade a day on average, and 5 on days with a trade

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67% of funds exhibit a negative alpha

# Portfolio Characteristics

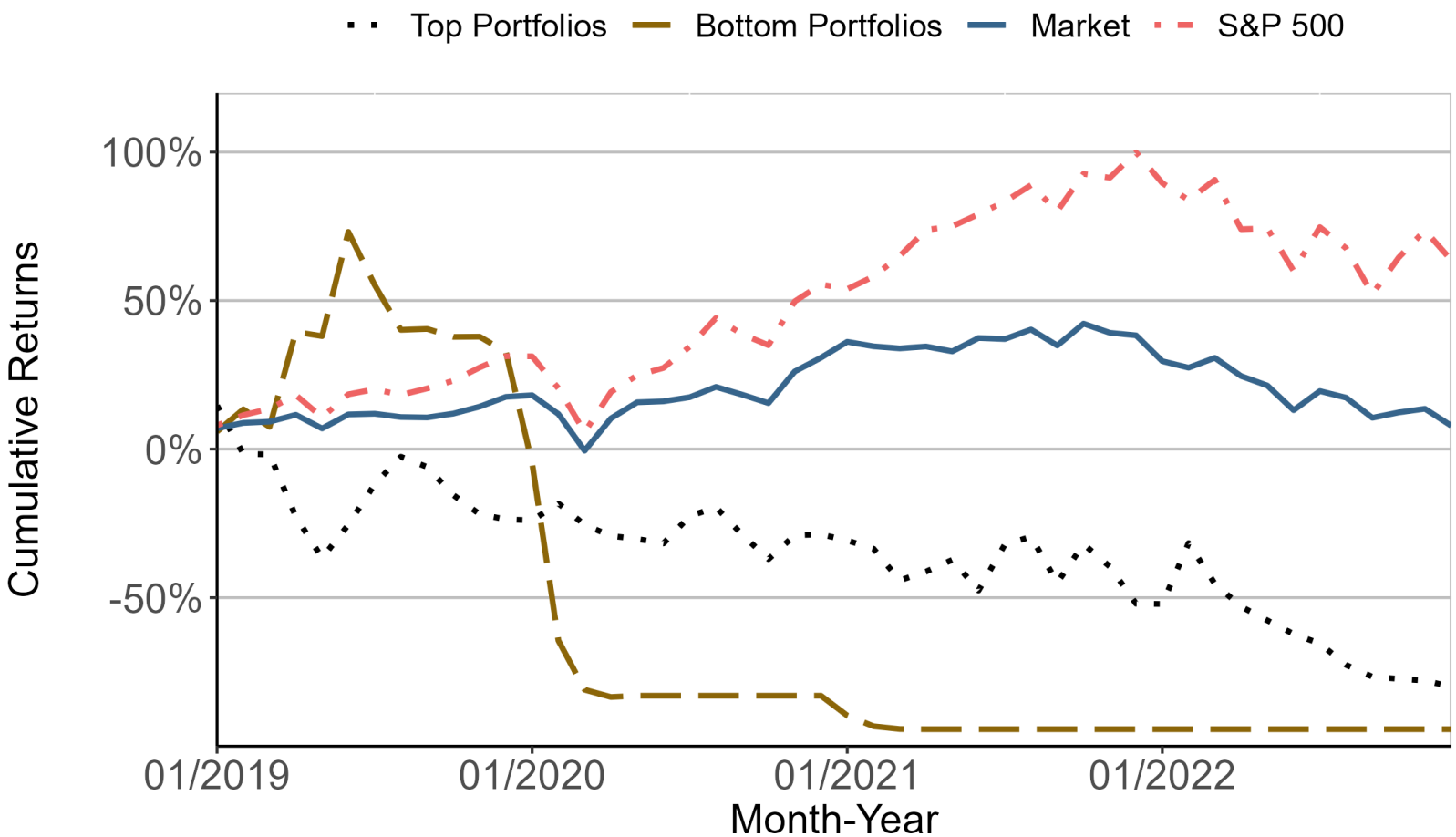
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About 30 bonuses a year on average (median 20)

# Return Benchmarking

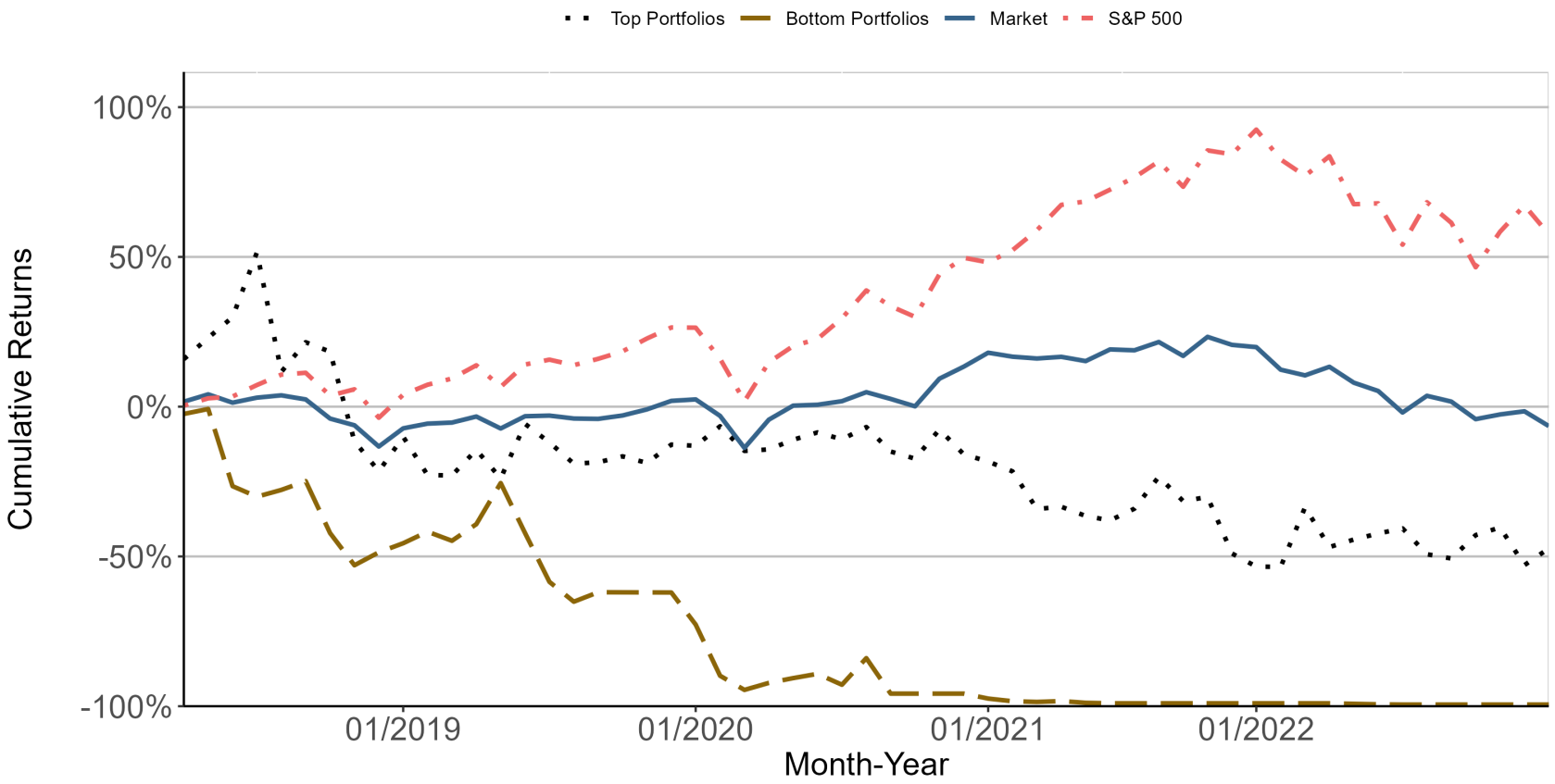
- Exercise
  - Rank funds by average return over the prior period
  - Invest in the ranked funds and hold for one period
  - Re-rank and rebalance either yearly, quarterly, or monthly
- Portfolios compared
  - **Top Portfolios:** 10 funds with the highest prior-period return
  - **Bottom Portfolios:** 10 funds with the lowest prior-period return
  - **Market:** equal-weighted, every fund active that period
  - **S&P 500:** benchmark, since 68% of funds trade US assets

# Yearly Rebalancing



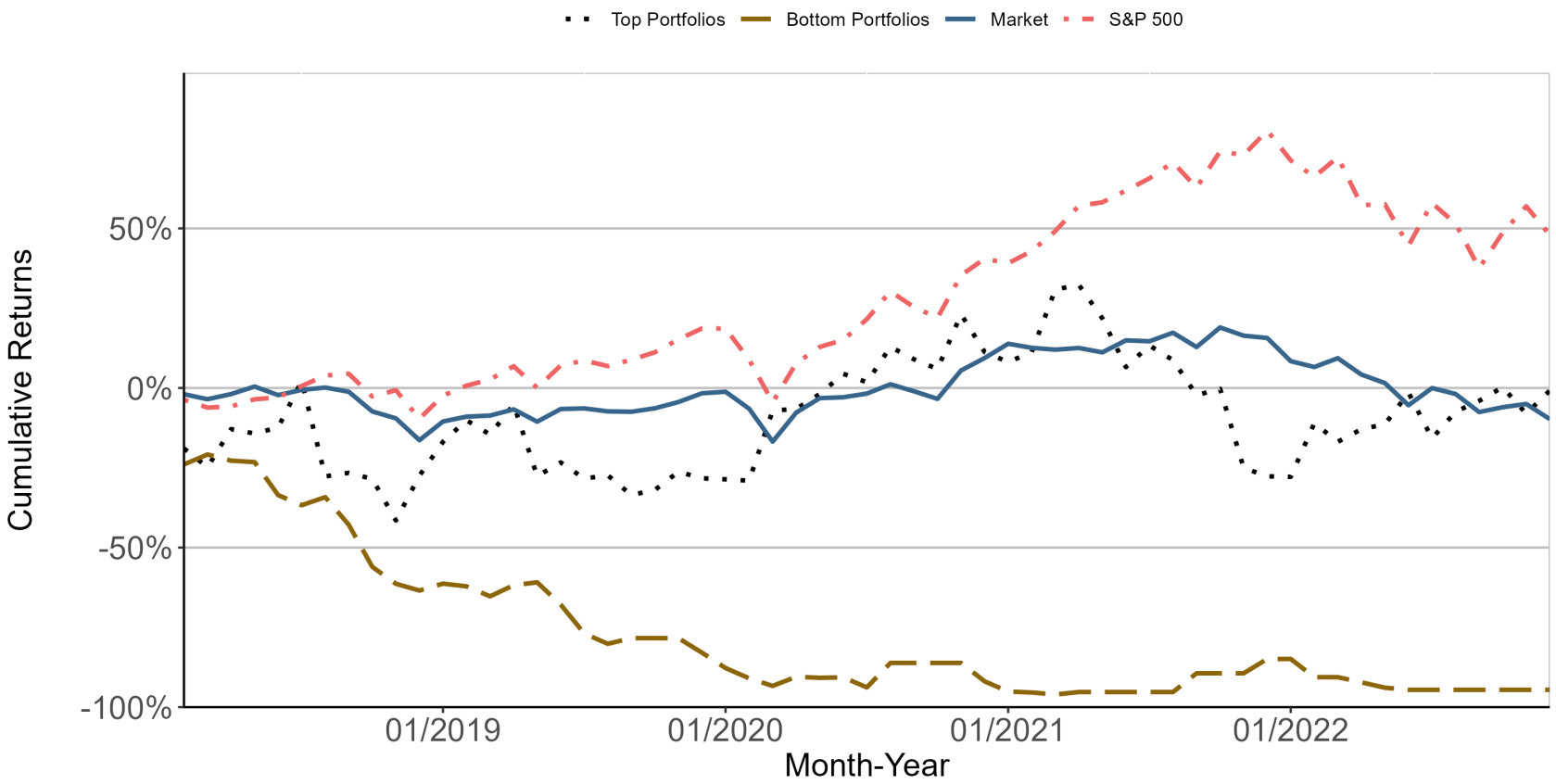

Both the top and bottom funds yield almost complete losses

# Quarterly Rebalancing




Slight improvement for the top funds, which only lose about 50%

# Monthly Rebalancing

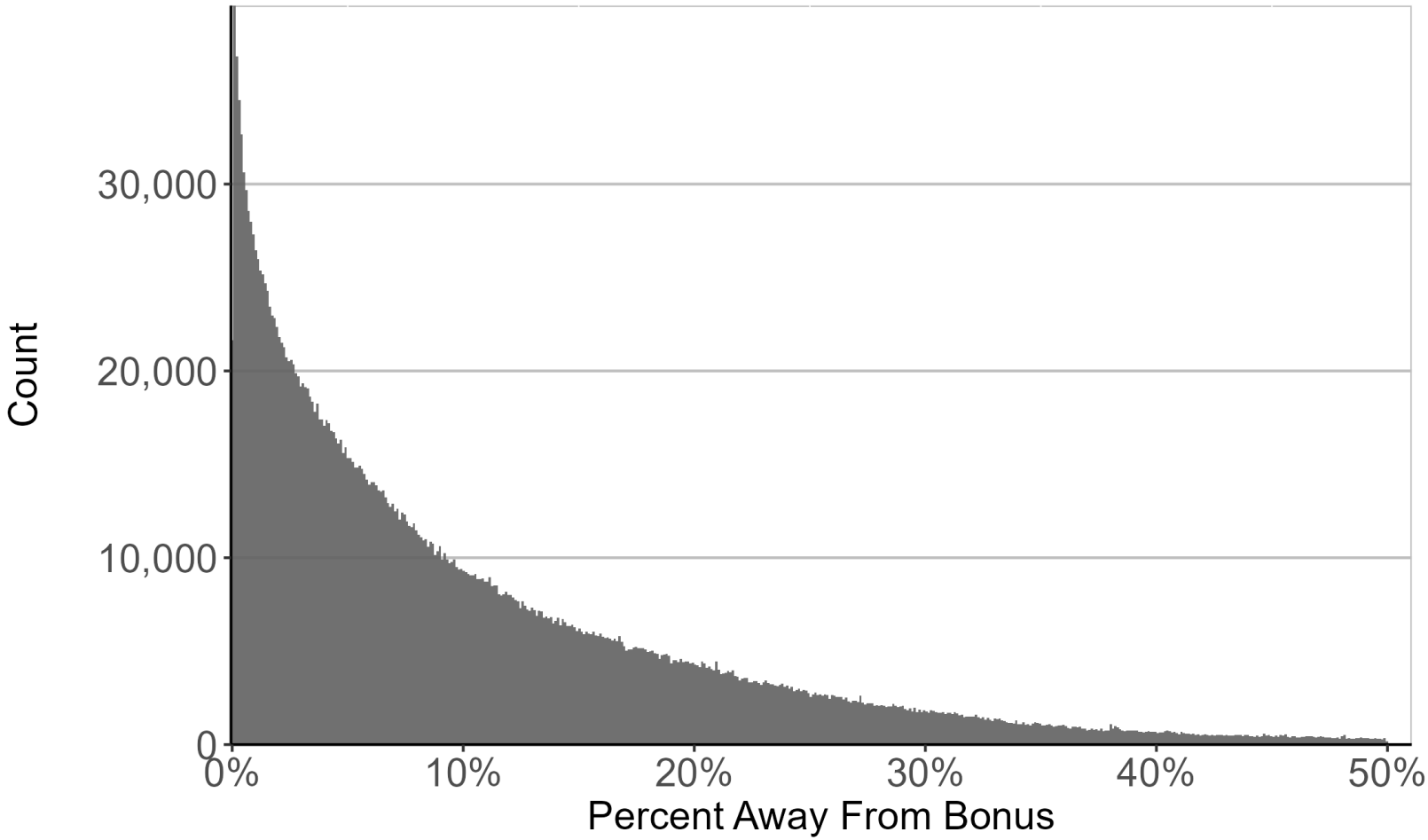



Top performers break even, but no consistent performance

# Why Doesn't Performance Persist?

- In line with traders favoring short-term strategies, rather than buy and hold
  - Most traded: leveraged ETFs on market indices, and stocks like Apple, Tesla, Amazon, and Nvidia
- Daily HWM could encourage this
- **How close are traders to a bonus on a typical day?**

# Distribution of Distance to Bonus

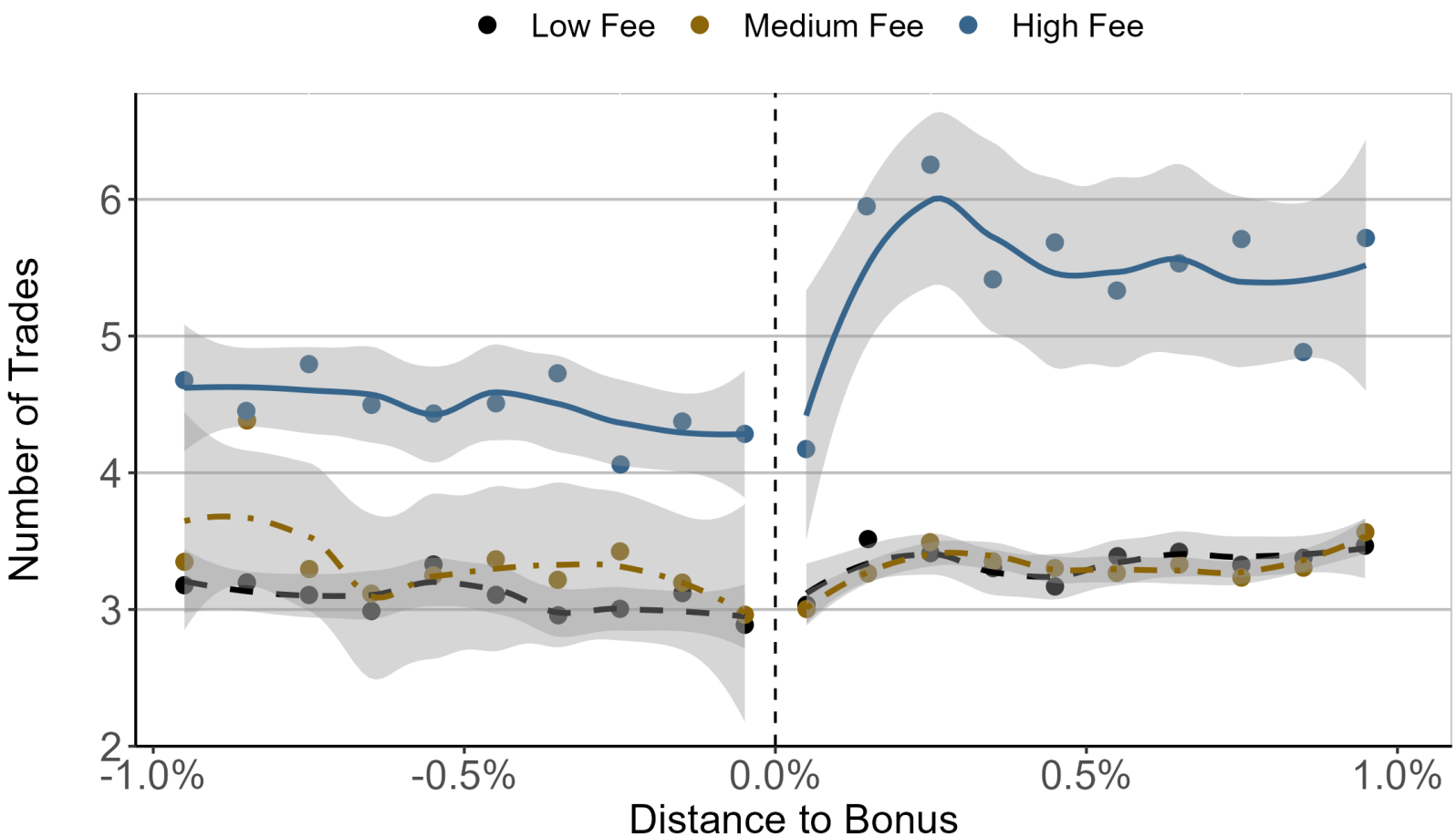



39% of all trader-days are within 5% of the bonus threshold

# Is the Bonus Salient?

- Regression discontinuity around the bonus threshold
- Compare trader-days just above vs. just below the threshold (within $\pm 1\%$)
- Outcomes over the prior 3 trading days (traders average about 1 trade a day):
  - Number of trades
  - Whether any trade was made
- Split by bonus rate (fixed at creation):
  - Low: $\leq 7.5\%$ (1,881 traders)
  - Medium: 7.5–15% (2,375 traders)
  - High: $> 15\%$ (891 traders)

# RDD Around Bonus By Group



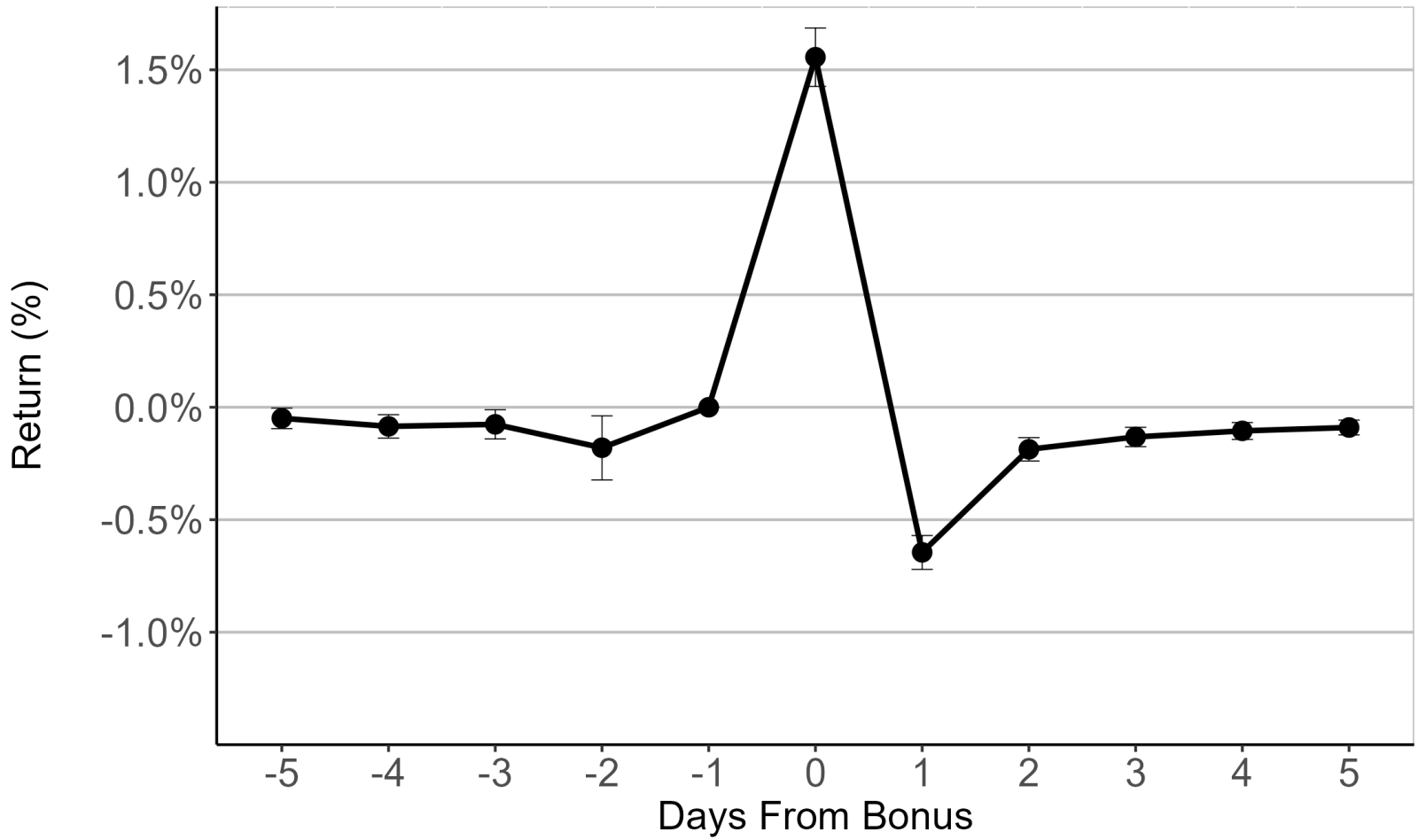

Higher bonus motivates greater trading around the HWM

# Differences-in-Differences Estimator

- Treatment = earning a bonus, which is
  - **Non-absorbing:** the HWM resets every January 1st
  - **Non-binary:** traders earn many bonuses a year
- Two-way fixed effects is biased with heterogeneous, staggered effects (Callaway and Sant'Anna 2021; Sun and Abraham 2021)
- Estimator of de Chaisemartin and D'Haultfoeuille (2026): change from the day before a bonus, vs. funds with the same treatment history not yet treated

$$DID_{g,\ell} = Y_{g,F_{g-1}+\ell} - Y_{g,F_{g-1}} - \frac{1}{N_{g,F_{g-1}+\ell}} \sum_{g':D_{g',1}=D_{g,1},F_{g'}>F_{g-1}+\ell} \left[Y_{g',F_{g-1}+\ell} - Y_{g',F_{g-1}}\right]$$

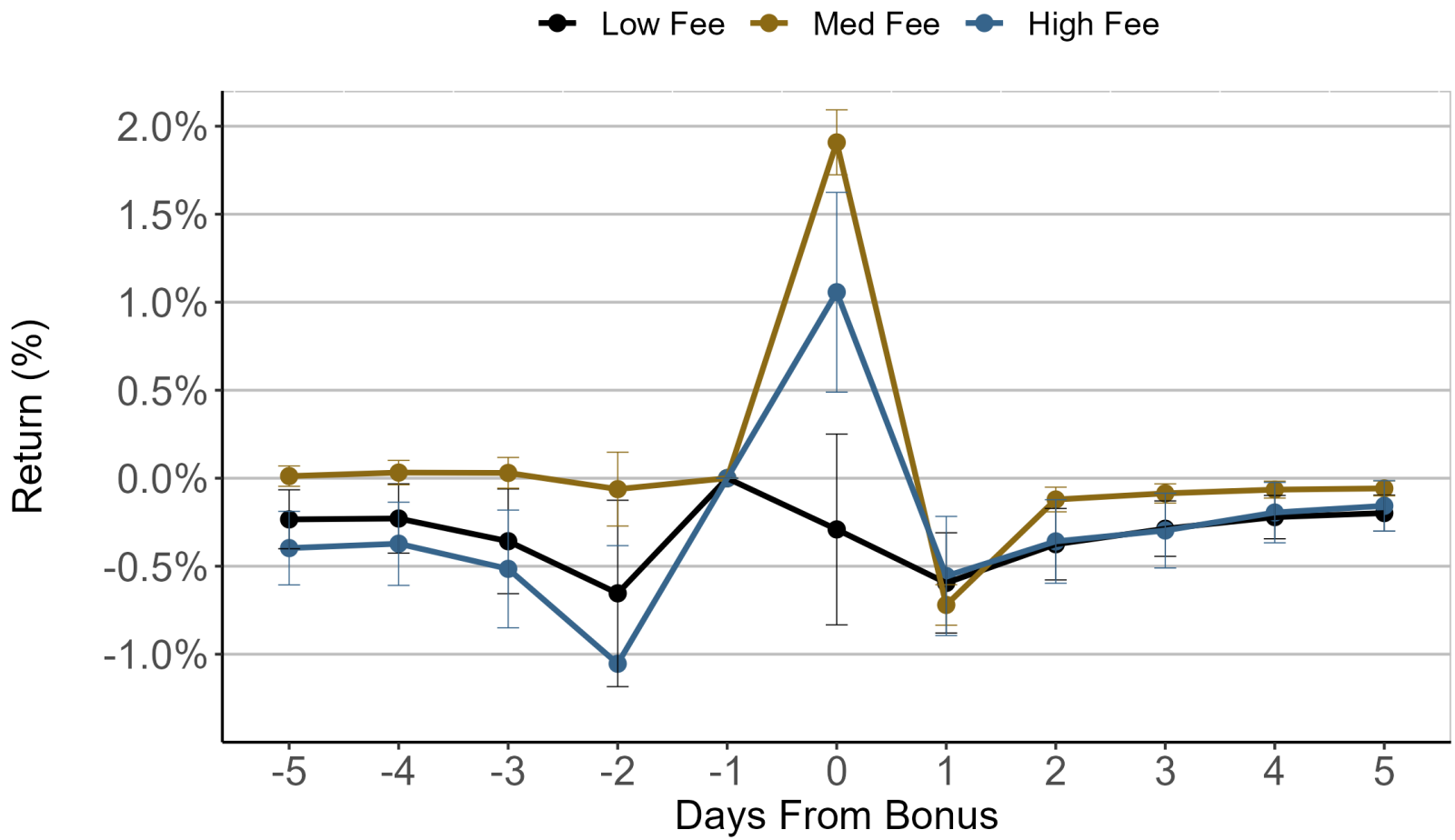
# Returns Around Bonuses




Traders earn a 1.6 pp higher return on the bonus day

▸ Regression Table

# Returns Around Bonuses By Performance Fee



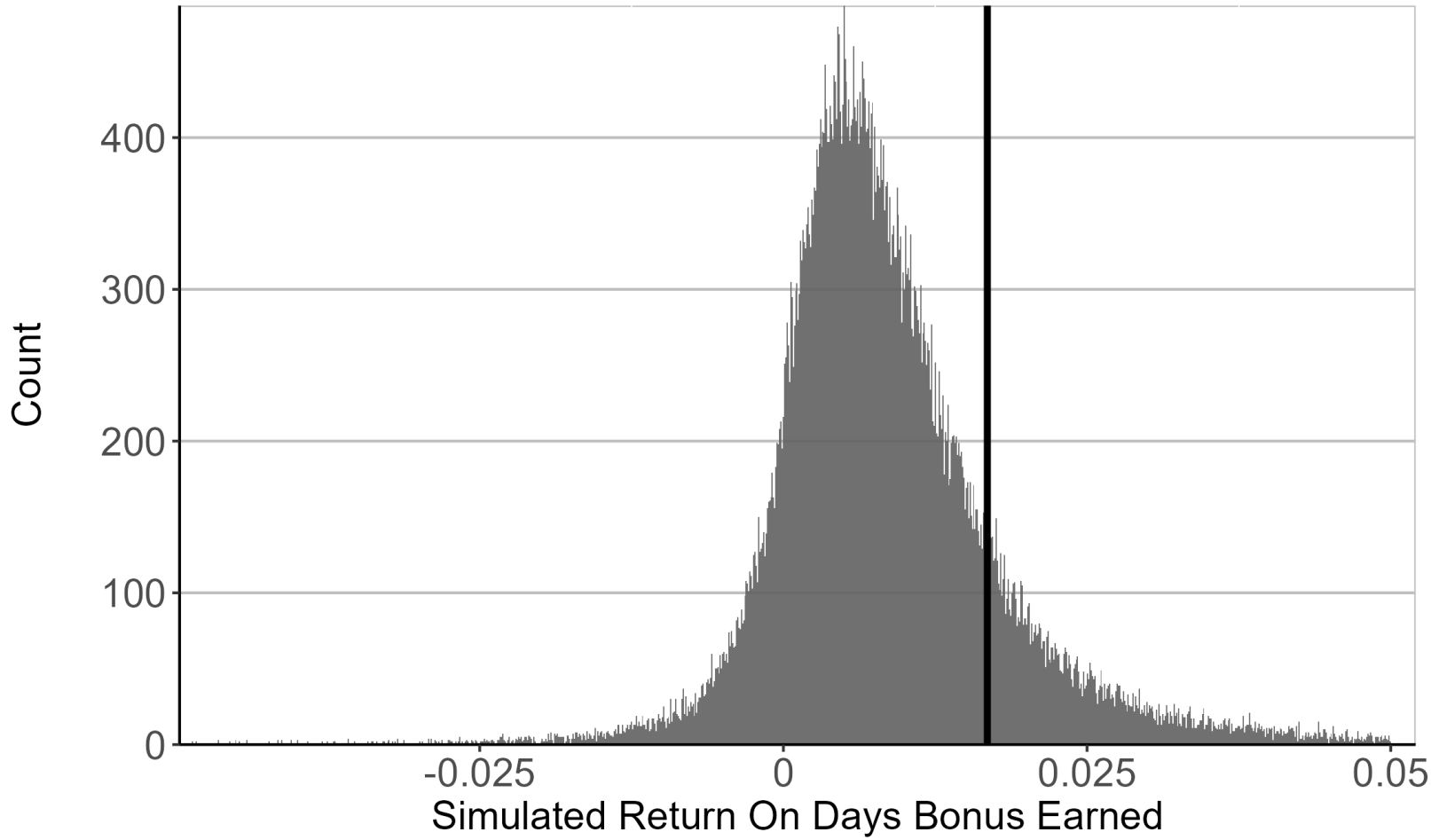

Higher fee traders have higher returns, consistent with incentives

▸ Number of Trades

# Luck or Strategy?

- Returns must be positive on a bonus day, so how big is a 1.6 pp jump?
- Luck vs. skill bootstrap (Kosowski et al. 2006; Fama and French 2010)
- For each bonus day:
  - Keep the fund's portfolio weights from the prior day
  - Draw that day's asset returns at random from the assets held by all funds crossing that day
  - Repeat 10,000 times
- Compare the simulated returns to the actual bonus-day return

# Bootstrap Return Distribution

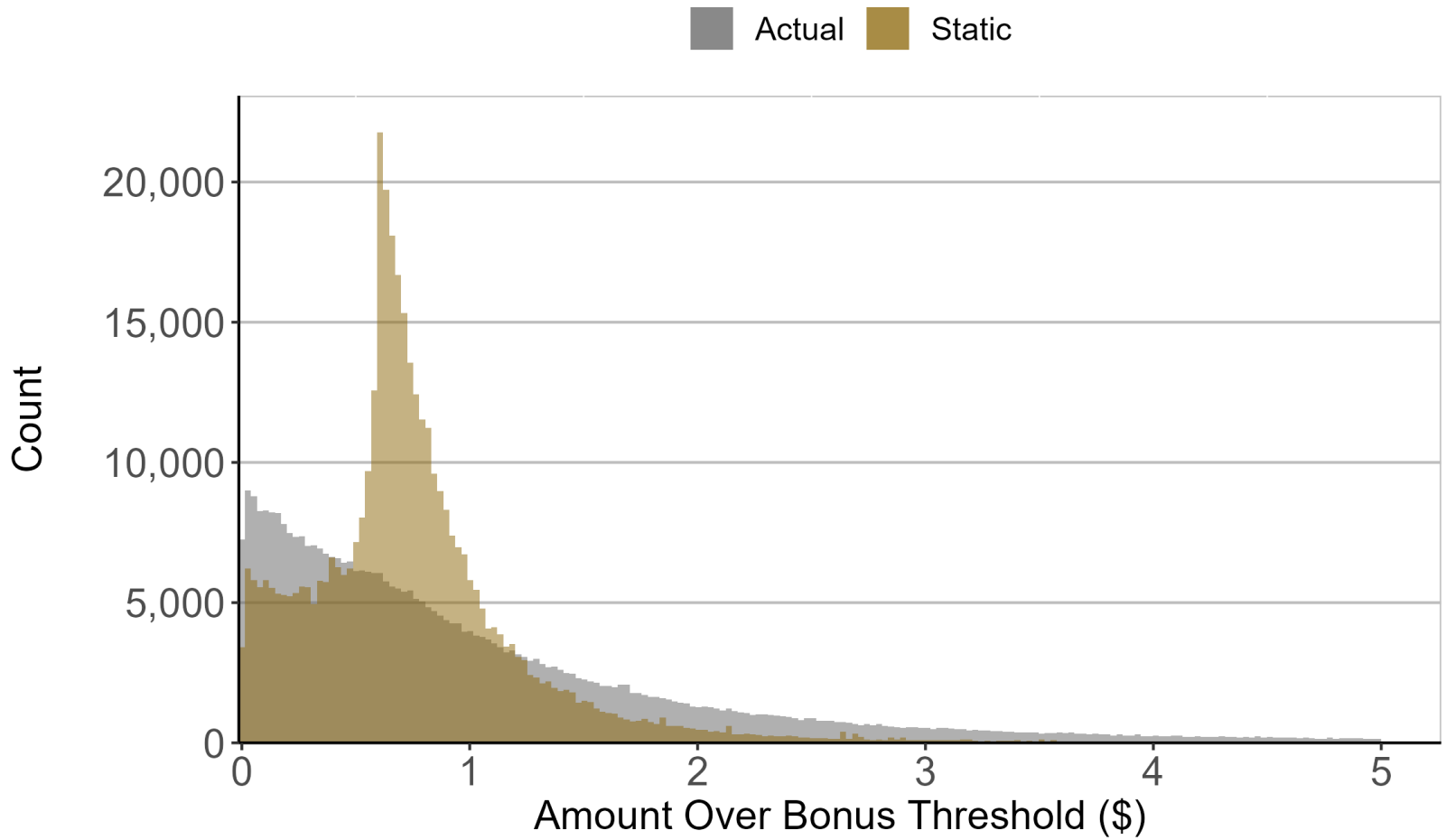



Observed 1.6% return is larger than 82% of simulations

# Static Portfolio Counterfactual

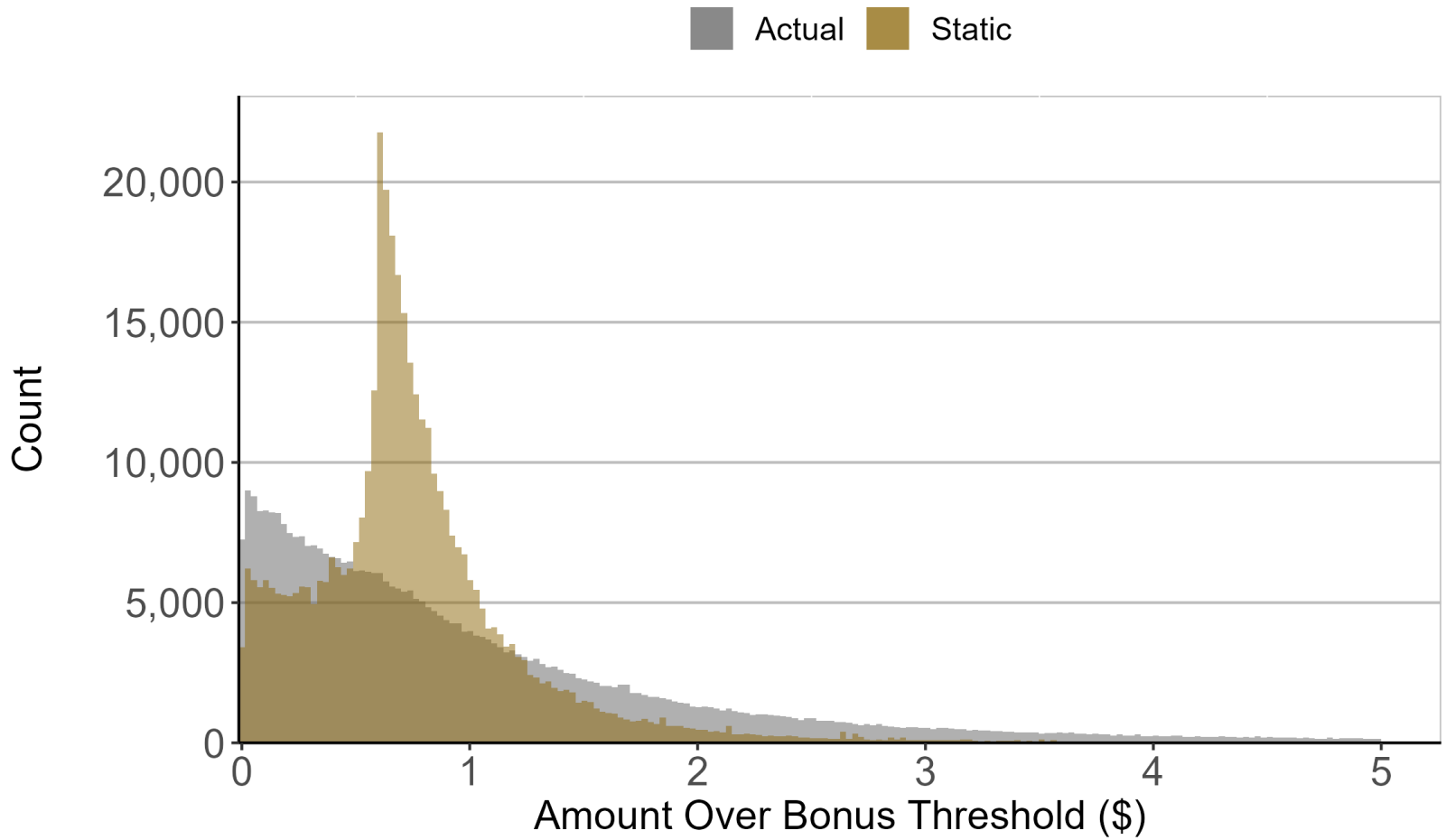
- What if traders did not trade at all?
- Freeze each fund's holdings on February 1st and hold them for a year
- Cumulative returns are similar and FF3 alpha is higher under a static setting for 63% of funds
- But is this optimal for bonus achievement?

# Distance Over Bonus Threshold




Under the static trading strategy 57% of fund-years achieve more bonuses

# Distance Over Bonus Threshold




Long right tail of actual crossings, consistent with HWM incentive

# Summary of Additional Results

- **Mechanism** ▸ Details
  - Assets traded on bonus days had higher past 10-day returns (sales 7.4% vs. 4.8% for purchases)
  - Sold assets went on to earn more: 0.69 pp higher future returns than purchases
  - Sales jump from about 40% to 48% of trades on bonus days
- **Investors (new data, preliminary)** ▸ Details
  - Certificate trades rise about 25% on bonus days (top 20% most-traded)
  - Volume rises too, but the estimate is noisy
- **January reset** ▸ Details
  - All HWMs reset on January 1st, so bonus timing comes from the calendar
  - Similar return jump on bonus days as the full sample

# Conclusion

- Spike in returns corresponds with selling high-return assets and forgoing potential further gains
- High-frequency incentives can be counterproductive for retail investors
  - Exacerbating short-term biases and undermining sustainable performance
- Highlights importance of incentive frequency and interaction with agent type

# Thanks!

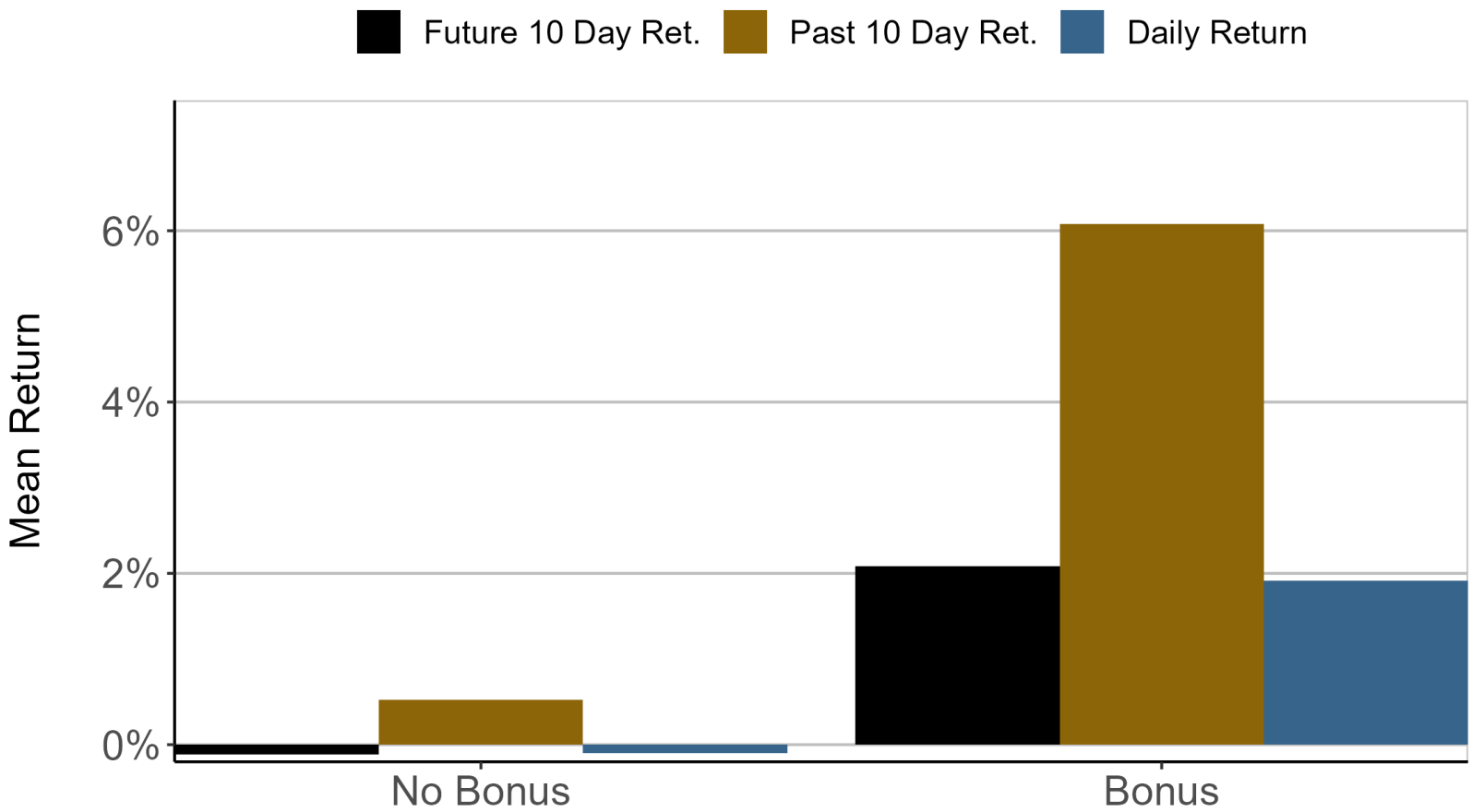
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# Mechanism: What Do Traders Trade?

- Use transaction-level data
- Compare assets bought and sold on bonus vs. non-bonus days
- Look at each asset's past 10-day, same-day, and next 10-day return

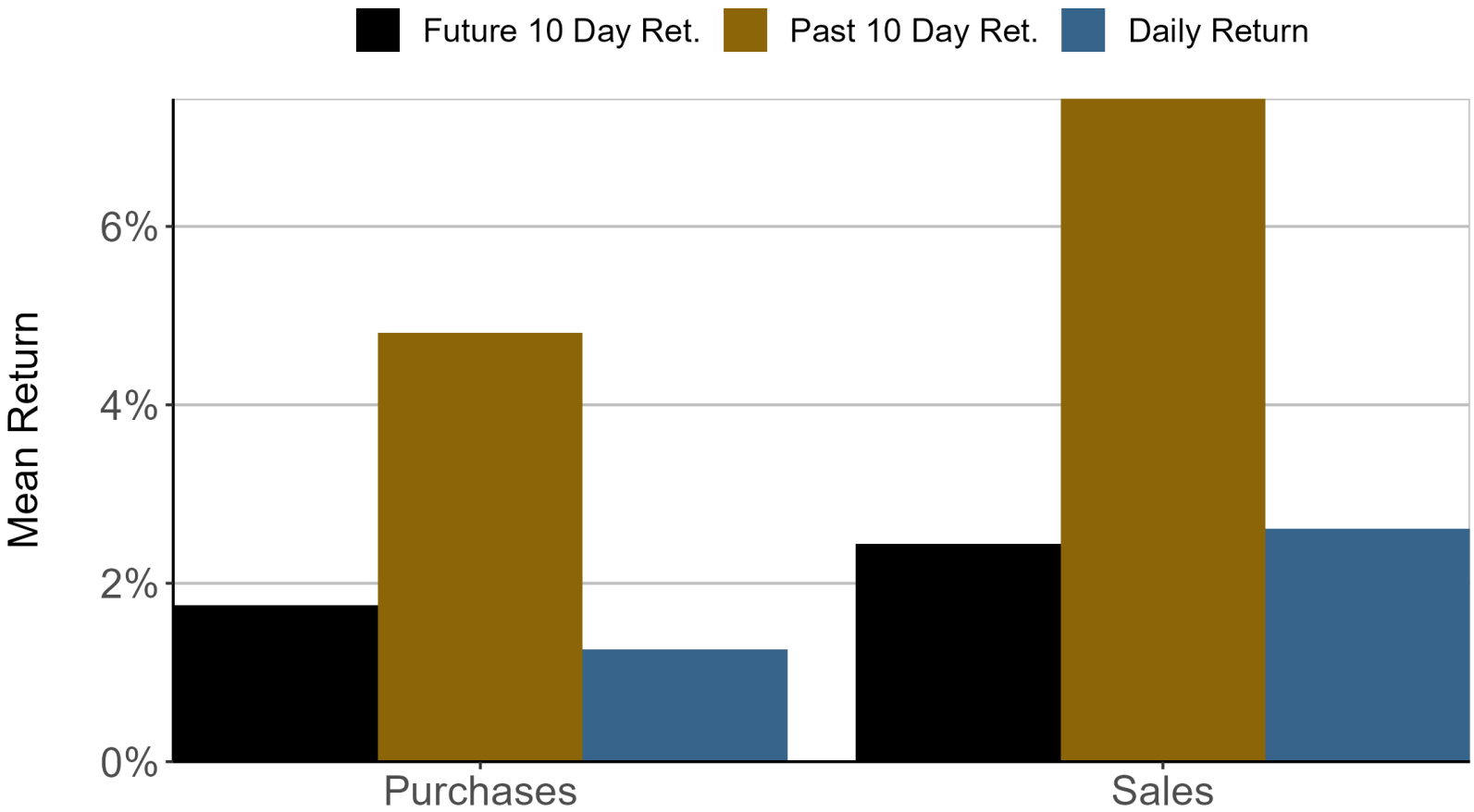
# Return Patterns: Bonus vs. Non-Bonus Days




Assets traded on bonus days had higher past returns

▸ Sales Table ▸ Purchases Table

# Return Patterns: Purchases vs. Sales on Bonus Days



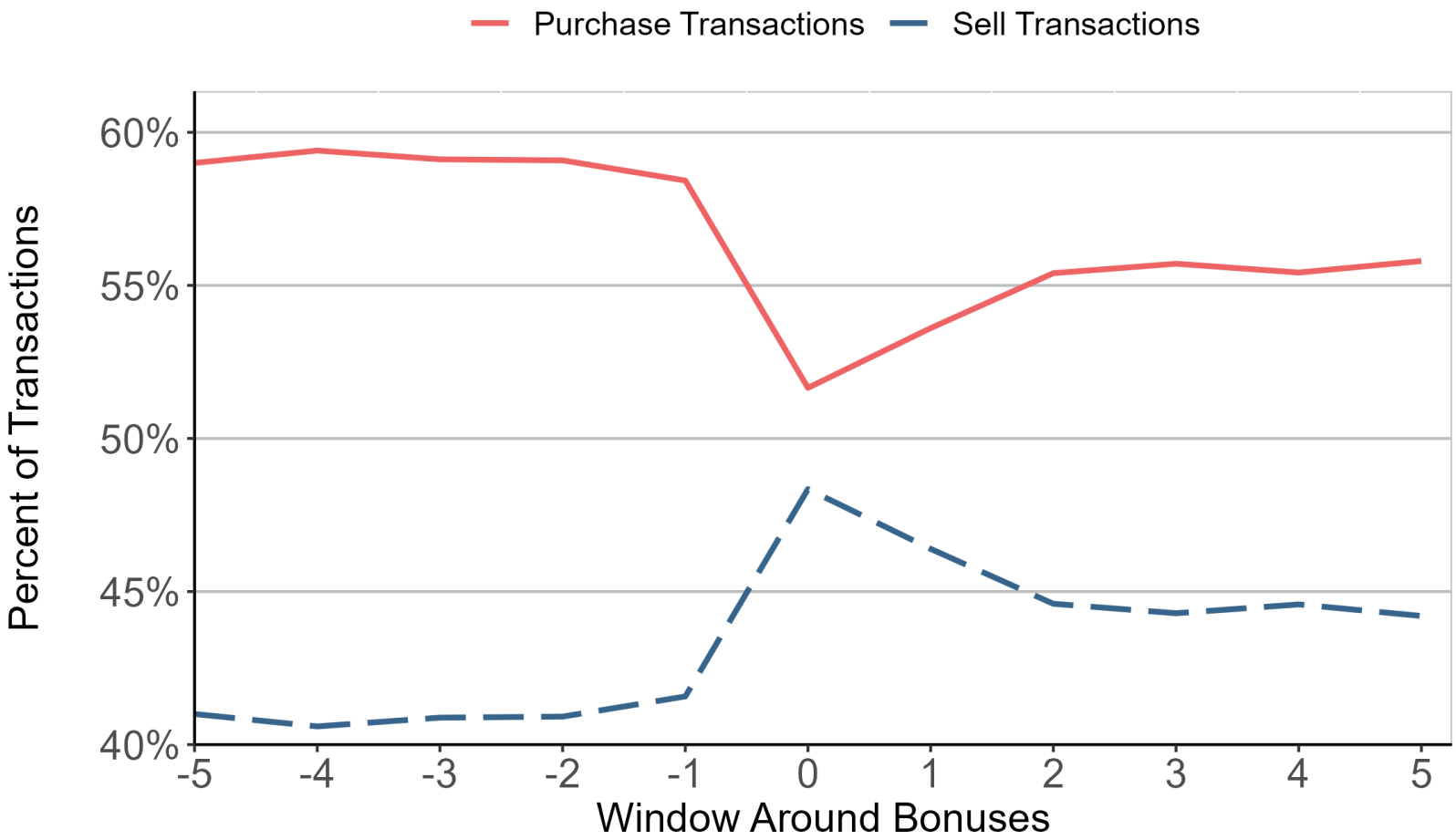

Future returns are larger for assets that traders sold on bonus days

▸ Sales Table ▸ Purchases Table

# Selling to Lock In the Bonus

- Are traders selling recent winners to secure the bonus?
- Share of transactions that are sales, in the 10-day window around a bonus

# Buying vs. Selling Activity



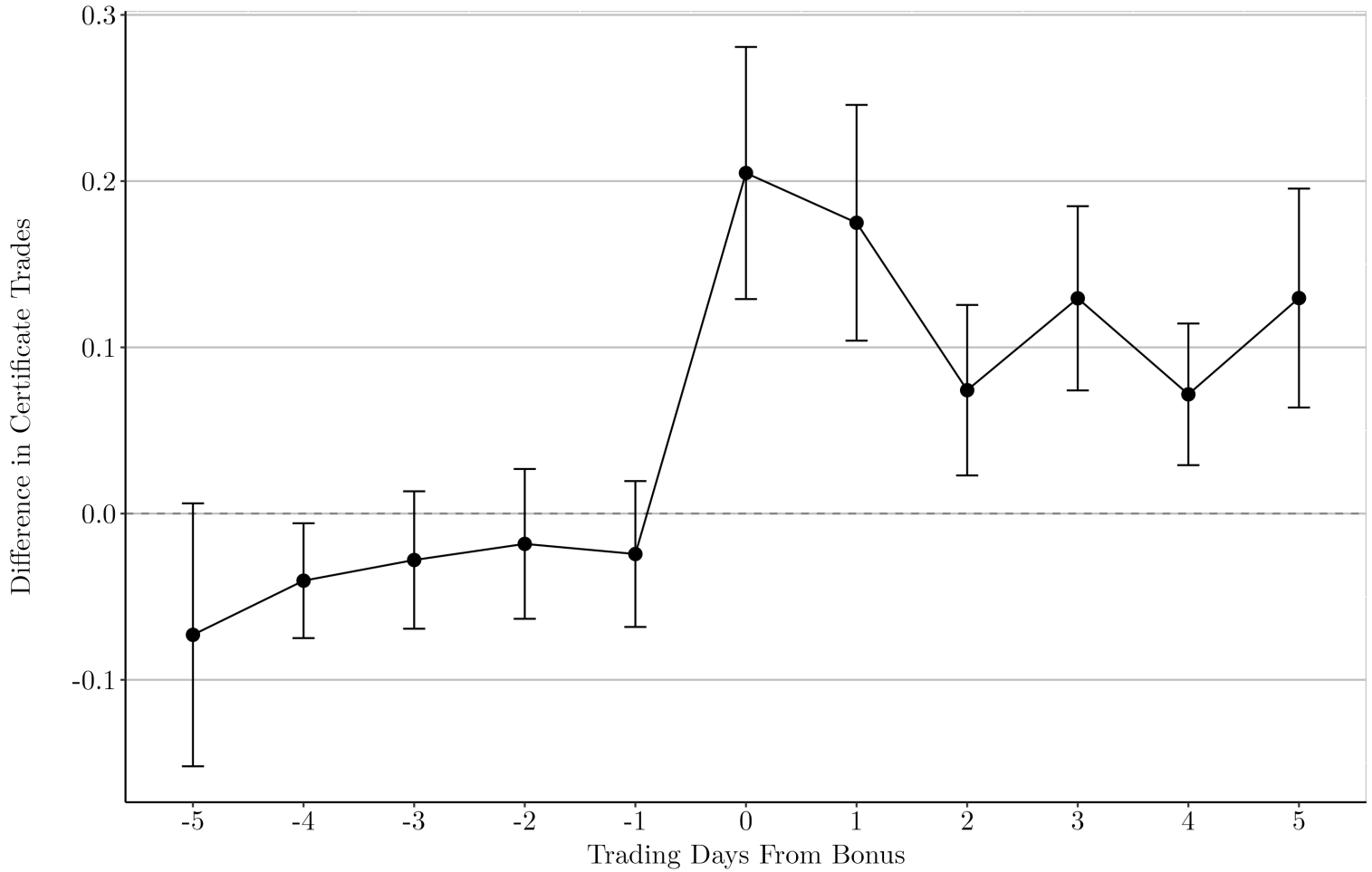

Consistent with selling recent winners to lock in the close price

Back to Summary

# New Data: How Investors Trade the Certificates

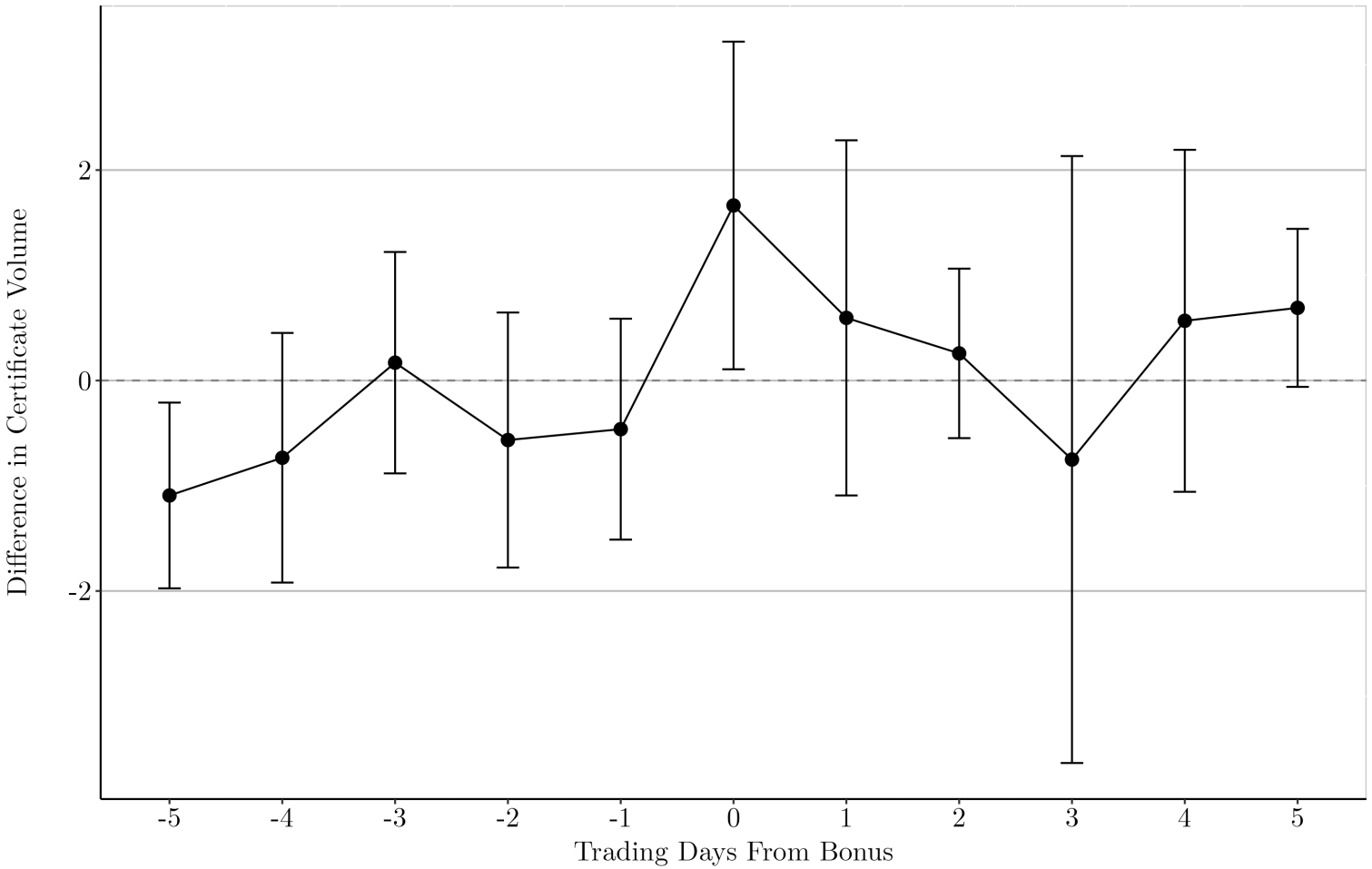
- Newly collected certificate trading data (preliminary results)
- Do investors respond to bonus days?
- Event study of certificate trades and volume around bonuses, in the most-traded certificates

# Certificate Trades Around Bonuses: Top 20% Most-Traded




∼25% of the 0.82 daily mean for these certificates

# Certificate Volume Around Bonuses



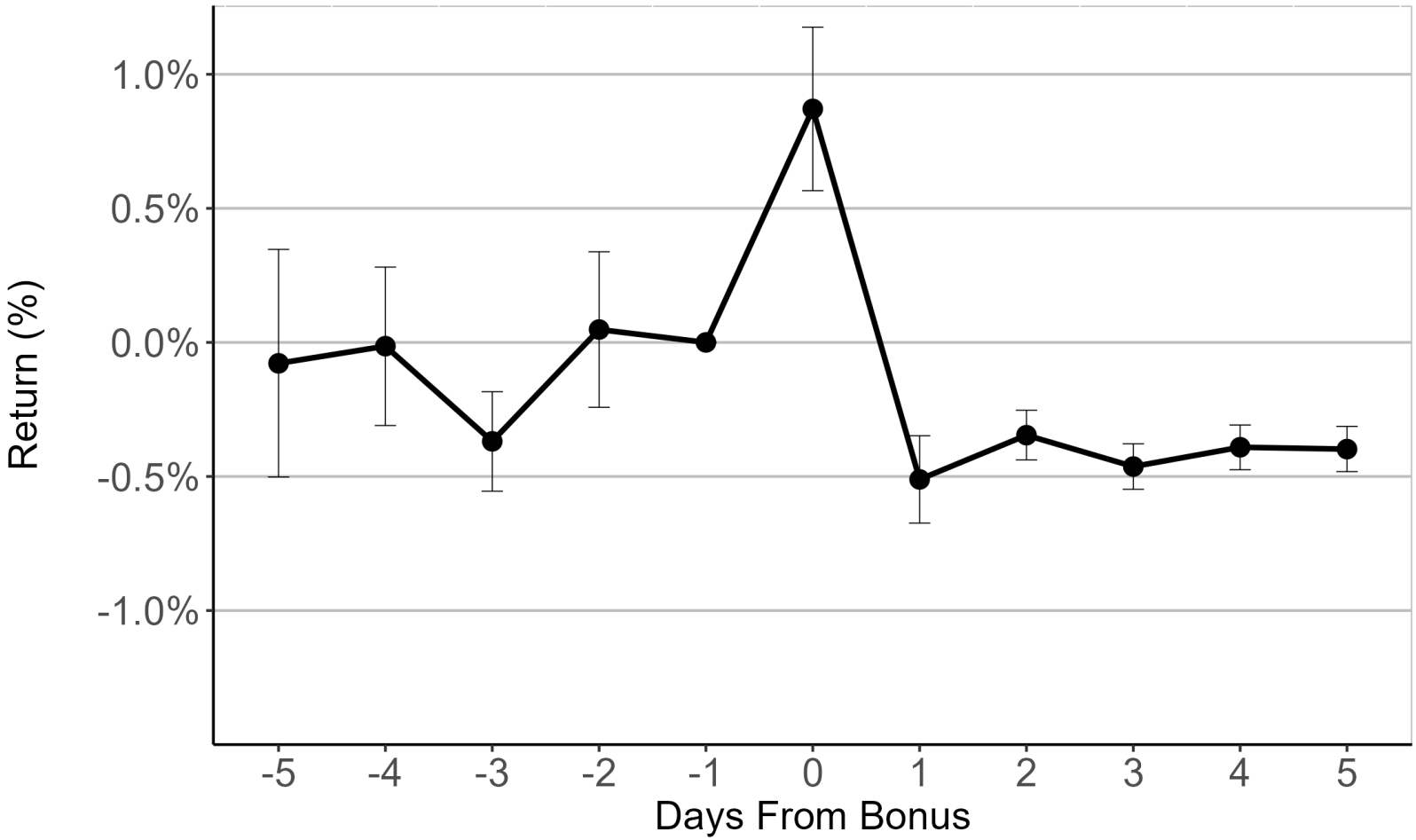

∼23% of the 7.29 daily mean for these certificates

Back to Summary

# January Reset

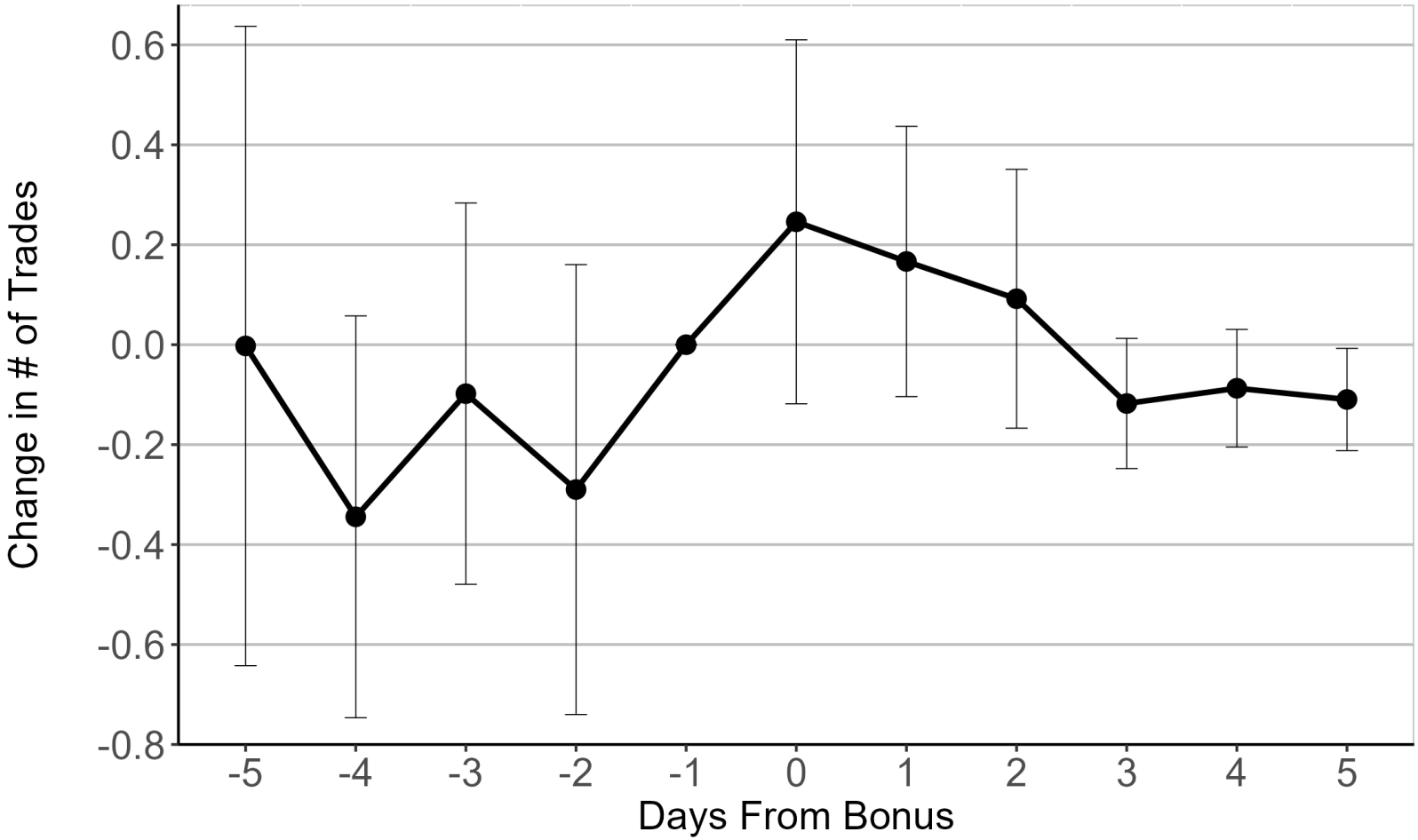
- Concern: more skilled traders may cross earlier or more often, and crossing depends on past performance
- Every January 1st all HWMs reset to that day's close
  - More easily achievable bonus allows us to isolate its effect
- Re-run the DiD on January only, first 3 bonuses per trader (437,046 observations)

# January: Returns Around Bonuses




Similar pattern to the whole sample

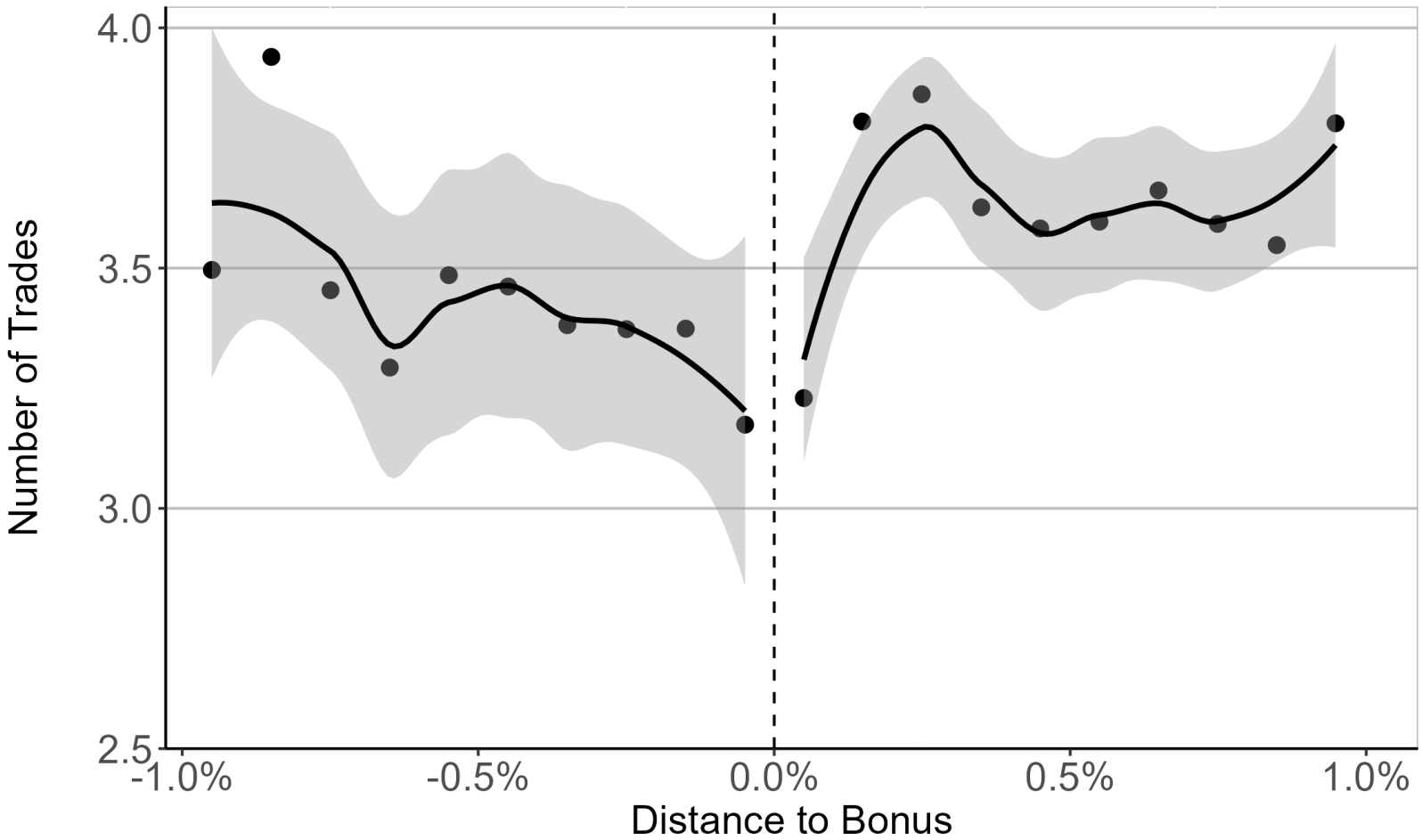
# January: Number of Trades Around Bonuses




Noisy, yet positive, estimate of trades

Back to Summary

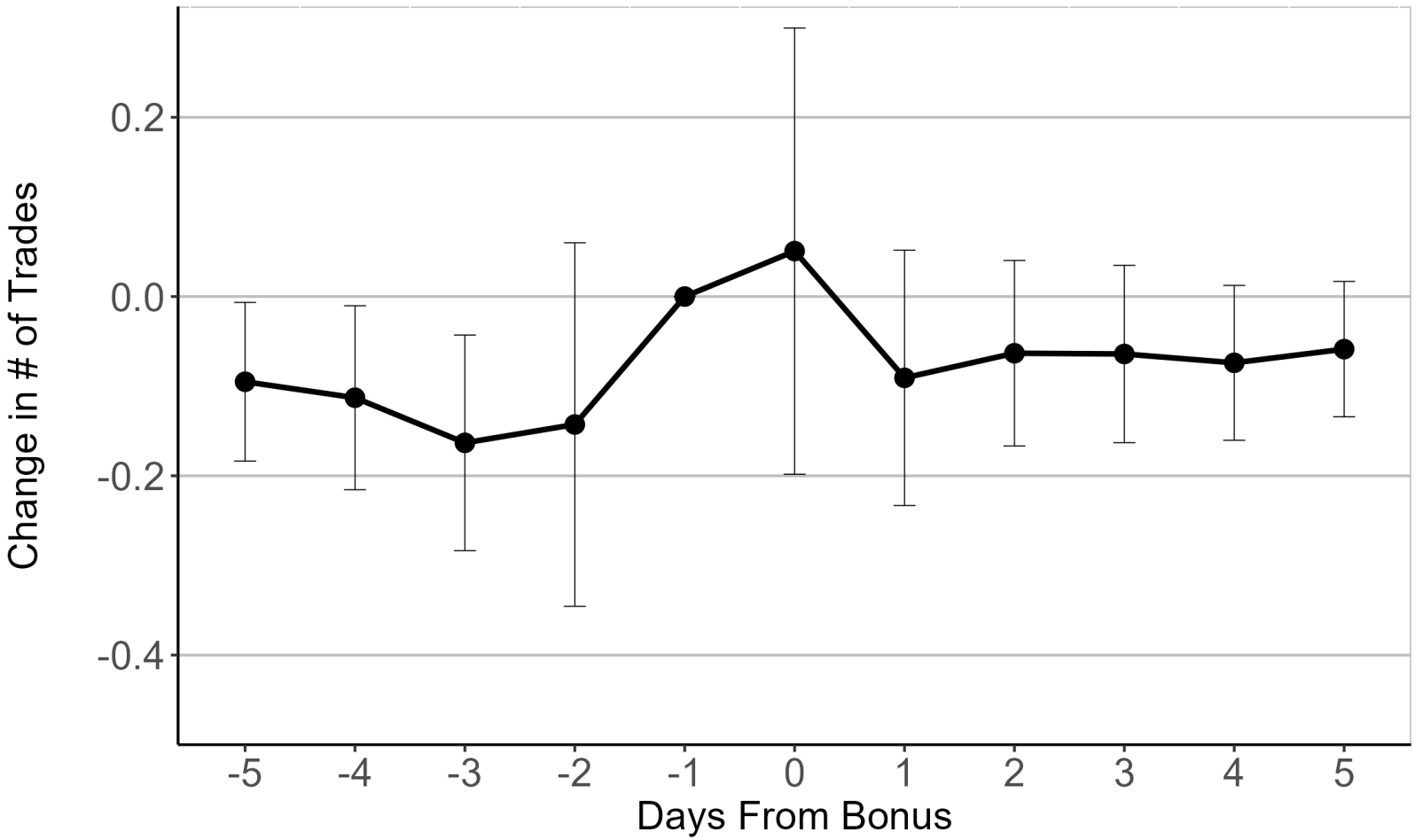
# RDD Around Bonus




Intensive: 0.31 trades (9% of mean) | Extensive: odds increase 15%

▸ Regression Table

# Number of Trades Around Bonuses




No significant increase in trades placed

▸ Regression Table

# Who Pushes Hardest for the Bonus?

- If the bonus drives the return jump, traders with the most to gain should push hardest
- A higher bonus rate means a larger payout from crossing the HWM
- Test: re-run the DiD around bonuses separately for low, medium, and high fee traders
- Prediction: the bonus-day return jump shows up in the higher-fee groups

# Back of the Envelope

- Is chasing the bonus worth it for the trader?
- **Gain:** the bonus as a share of fund value, $Bonus_{i,t}/Price_{i,t}$
- **Cost:** abnormal return, the next 10-day return after the bonus minus the 10-day return just before it

$$Net_{i,t} = \frac{Bonus_{i,t}}{Price_{i,t}} + \left( R_{i,t}^{+10} - R_{i,t}^{-10} \right)$$

- $Net_{i,t} < 0$: the push for the bonus cost more than it paid
- Average net loss of 2.5% of fund value; median of $-1.4\%$
  - Statistically different from zero; robust to window length and to overlapping bonus windows

# Social Trading (Wikifolios)

- Individuals register and build a virtual portfolio using over ∼600,000 financial products, including
  - Stocks
  - Exchange-traded products
  - Funds (real estate, bonds, equities, money market, or mixed funds)
  - Investment certificates (index certificates, bonus certificates, discount certificates, and reverse convertibles)
  - Leverage products
- Traders trade virtually, but prices reflect real-time bid/ask quotes
- An index is created to replicate the virtual portfolio's performance 1:1

# Starting a Wikifolio

- Any registered user can create a wikifolio: a virtual portfolio plus a written trading idea (at least 200 characters)
- The trader chooses the investment universe the wikifolio can trade
- **Test** status: private, only the trader can see it
- To publish: trader profile, title, idea, at least one trade, and verified email
- **Published** status: searchable, users can add it to their watchlist and follow every trade

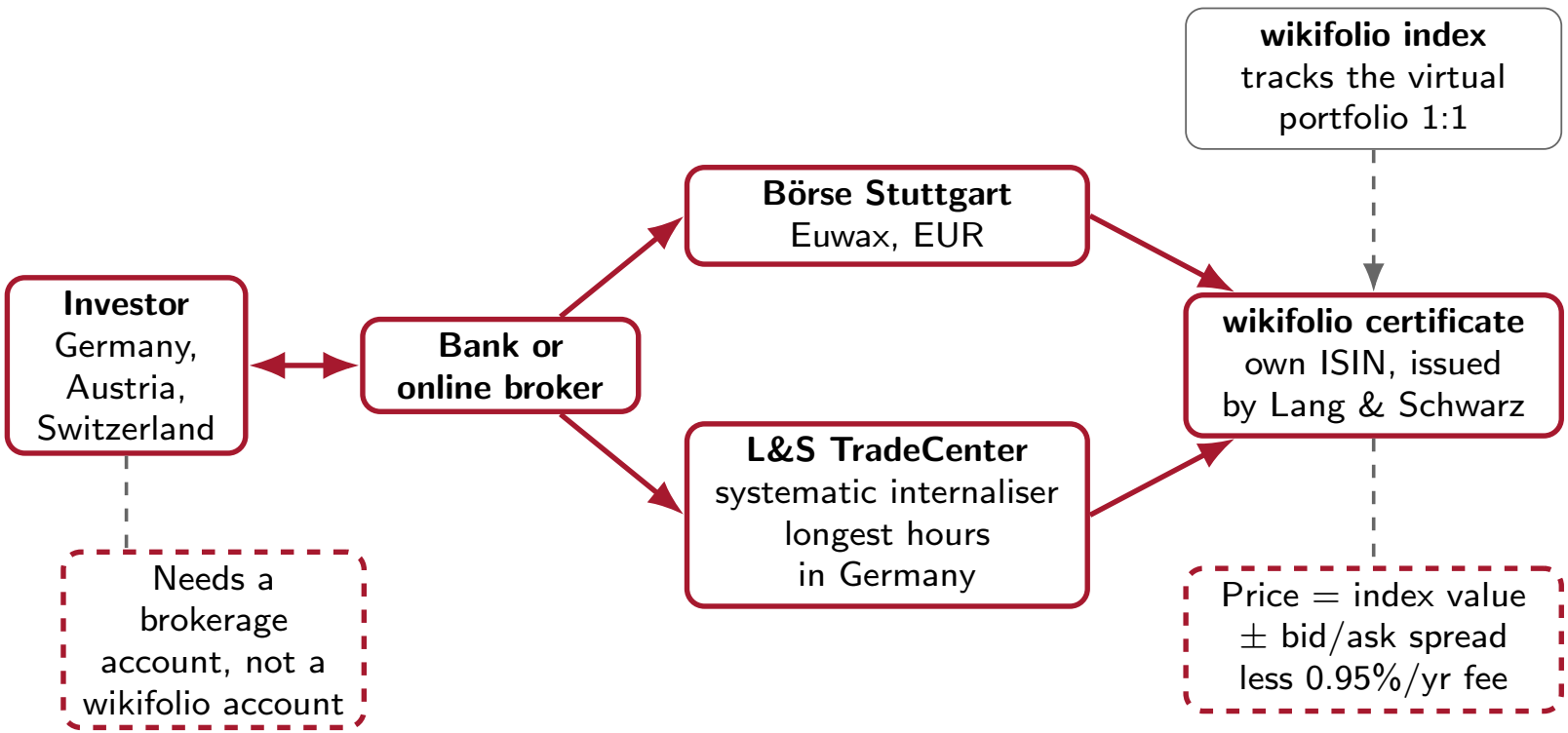
# Becoming Investable

- At least 10 watchlistings and the wikifolio is at least 21 days old
- Legitimation: ID, address, and phone number (18 or older)
- Issuance call with wikifolio about the trading idea and strategy
- Lang & Schwarz issues an open-ended index certificate with its own ISIN that tracks the wikifolio 1:1
- Once issued, the investment universe is fixed

# Investing in Wikifolio Certificates

- Investors buy the certificate by its ISIN through their own bank or online broker, with no wikifolio account needed
- The certificate is not traded on the Lang & Schwarz exchange, but on
  - Börse Stuttgart (Euwax, in EUR), or
  - Directly with Lang & Schwarz TradeCenter: a systematic internaliser, Germany's off-exchange venue with the longest trading hours
- Who can buy
  - Germany: every certificate is publicly offered
  - Austria: certificates of Austrian traders or with high demand
  - Switzerland: only stock-only wikifolios (BX Swiss, in CHF)
- Price tracks the wikifolio 1:1, with a bid/ask spread from its holdings and a 0.95% annual certificate fee charged daily

# How Investors Buy In



# Fees

- Traders earn no fee on assets under management, only performance bonuses
- Bonus rate of 5–30% of the gain above the high water mark (HWM), set by the trader at launch and fixed
- Earned only when the daily close sets a new annual high
- The HWM resets every January 1st

# Bonus Calculation

$$Bonus_{i,t} = \begin{cases} (Price_{i,t} - HWM_{i,t}) * Rate_i, & \text{if } Price_{i,t} > HWM_{i,t} \\ 0, & \text{otherwise} \end{cases}$$

# Sample High Water Mark

| Date | Close Price | High Water Mark | Bonus Earned? |
|---|---|---|---|
| 6/3/22 | 154 | 156 | No |
| 6/6/22 | 153 | 156 | No |
| 6/7/22 | 157 | 156 | Yes |
| 6/8/22 | 156 | 157 | No |
| 6/9/22 | 151 | 157 | No |
| 6/10/22 | 153 | 157 | No |
| 6/13/22 | 158 | 157 | Yes |
| 6/14/22 | 161 | 158 | Yes |
| 6/15/22 | 156 | 161 | No |
| 6/16/22 | 157 | 161 | No |

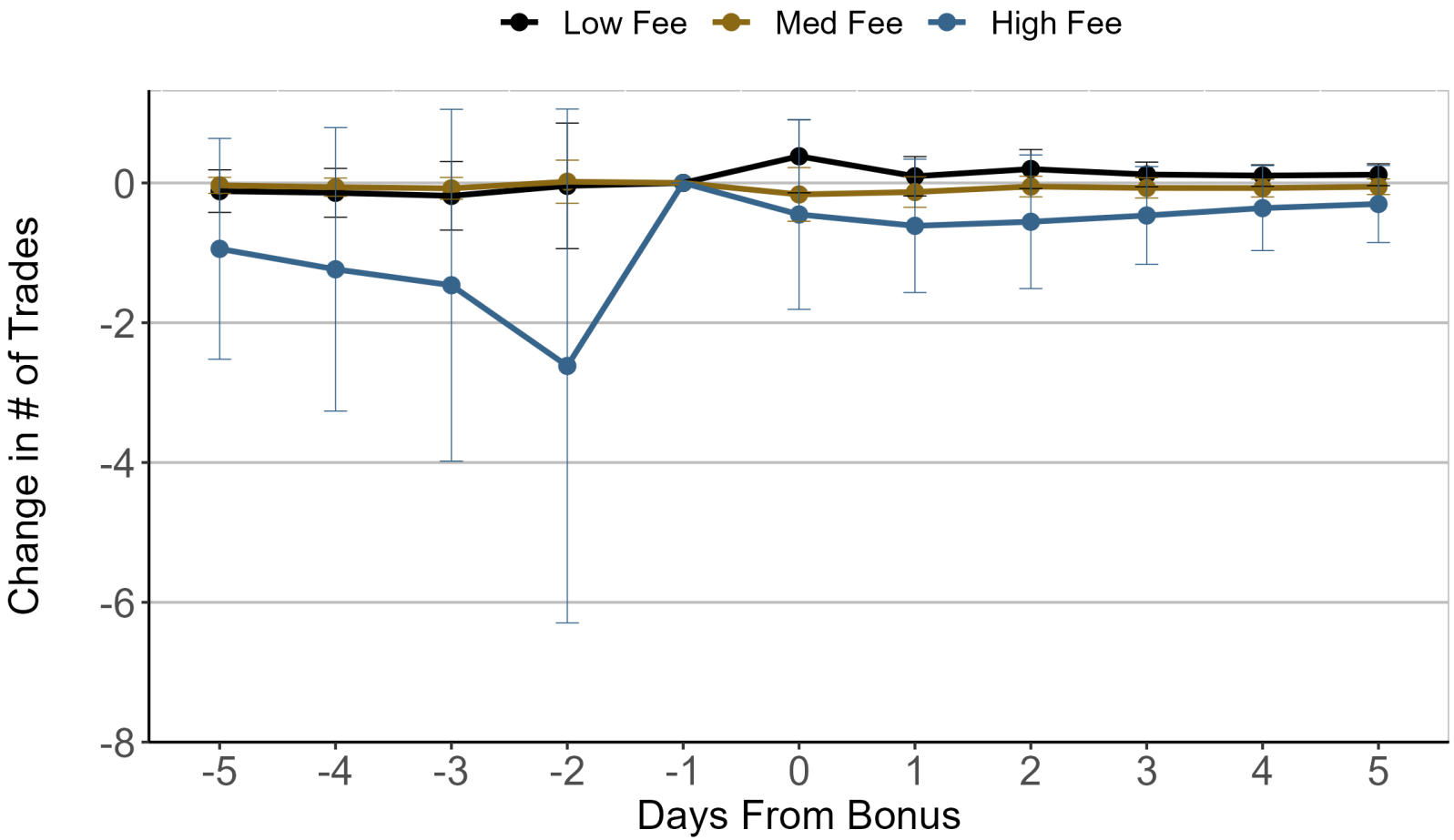
Bonus = (Close − HWM) × Rate; rate of 5–30% set by the trader; HWM resets January 1st

# Data

| | **Current** | **Extension** |
|---|---|---|
| Period | 2018–2022 | Through 2026 |
| Underlying returns | Daily returns from CRSP and LSEG | Tick-level exchange and OTC tape data, quotes and trades (finalizing collection) |

- **Filters:** drop closes below \$1, each fund's first month, January, and days over \$100 from a bonus; keep funds with 20+ trades a year and 6+ active months
- **Current sample:** 5,147 funds and 3.6 million fund-days (from 8,155 funds)

# Number of Trades Around Bonuses By Performance Fee




No difference across fee

Returns By Performance Fee

# RDD Around Bonus Achievement

| | Number of Trades | Probability of Trading |
|---|---|---|
| | (1) | (2) |
| Intercept | 3.237*** | -0.497*** |
| | (0.037) | (0.007) |
| I(Distance to Bonus $\geq$ 0) | 0.312*** | 0.140*** |
| | (0.051) | (0.009) |
| Distance to Bonus | -41.158*** | -17.497*** |
| | (6.745) | (1.244) |
| I(Distance to Bonus $\geq$ 0)$\times$ Distance to Bonus | 58.004*** | 17.217*** |
| | (9.845) | (1.811) |
| Observations | 626,187 | 626,187 |
| Adjusted $R^2$ | 0.0001 | |

◂ Number of Trades

# Bonus Achievement Difference in Differences

| | Returns | Number of Trades | Volatility |
|---|---|---|---|
| | (1) | (2) | (3) |
| Day -5 | -0.001** | -0.095** | 0.003 |
| | (0.0002) | (0.045) | (0.002) |
| Day -4 | -0.001*** | -0.113** | 0.004 |
| | (0.0002) | (0.052) | (0.002) |
| Day -3 | -0.001** | -0.163*** | 0.006 |
| | (0.0003) | (0.061) | (0.003) |
| Day -2 | -0.002** | -0.143 | -0.003 |
| | (0.001) | (0.103) | (0.004) |
| Day 0 | 0.016*** | 0.051 | 0.004 |
| | (0.001) | (0.127) | (0.004) |
| Day +1 | -0.006*** | -0.091 | 0.001 |
| | (0.0004) | (0.073) | (0.003) |
| Day +2 | -0.002*** | -0.063 | 0.0002 |
| | (0.0003) | (0.053) | (0.002) |
| Day +3 | -0.001*** | -0.064 | 0.001 |
| | (0.0002) | (0.051) | (0.003) |
| Day +4 | -0.001*** | -0.074 | -0.001 |
| | (0.0002) | (0.044) | (0.002) |
| Day +5 | -0.001*** | -0.059 | -0.001 |
| | (0.0002) | (0.039) | (0.002) |
| Observations | 3,250,182 | 3,250,182 | 3,250,182 |

◂ Returns ◂ Number of Trades

# Transactions When Bonus Achieved: Sales

| Dependent Variables: | Next 10 Day Returns | | Previous 10 Day Returns | |
|---|---|---|---|---|
| | (1) | (2) | (5) | (6) |
| Bonus Achieved | 0.012*** | 0.007*** | 0.045*** | 0.021*** |
| | (0.001) | (0.001) | (0.001) | (0.001) |
| Sale Transaction | -0.002* | -0.001 | 0.004** | 0.005*** |
| | (0.001) | (0.001) | (0.001) | (0.001) |
| Bonus Achieved × Sale Transaction | 0.005*** | 0.004*** | 0.022*** | 0.020*** |
| | (0.001) | (0.001) | (0.001) | (0.001) |
| Previous 10 Day Returns | 0.129*** | 0.121*** | | |
| | (0.003) | (0.003) | | |
| Daily Return | | | 0.022*** | 0.020*** |
| | | | (0.005) | (0.004) |
| *Fixed-effects* | | | | |
| Fund | Yes | Yes | Yes | Yes |
| Date | | Yes | | Yes |
| Observations | 3,620,016 | 3,620,016 | 3,635,974 | 3,635,974 |
| Adjusted $R^2$ | 0.026 | 0.021 | 0.025 | 0.011 |

◂ Bonus Vs Non Bonus Days ◂ Purchases Vs Sales

# Transactions When Bonus Achieved: Purchases

| Dependent Variables: | Next 10 Day Returns | | Previous 10 Day Returns | |
|---|---|---|---|---|
| | (3) | (4) | (7) | (8) |
| Bonus Achieved | 0.017*** | 0.011*** | 0.067*** | 0.041*** |
| | (0.001) | (0.001) | (0.002) | (0.001) |
| Buy Transaction | 0.002* | 0.000 | -0.004** | -0.005*** |
| | (0.001) | (0.001) | (0.001) | (0.001) |
| Bonus Achieved × Buy Transaction | -0.005*** | -0.004*** | -0.022*** | -0.020*** |
| | (0.001) | (0.001) | (0.001) | (0.001) |
| Previous 10 Day Returns | 0.129*** | 0.121*** | | |
| | (0.003) | (0.003) | | |
| Daily Return | | | 0.022*** | 0.020*** |
| | | | (0.005) | (0.004) |
| *Fixed-effects* | | | | |
| Fund | Yes | Yes | Yes | Yes |
| Date | | Yes | | Yes |
| Observations | 3,620,016 | 3,620,016 | 3,635,974 | 3,635,974 |
| Adjusted $R^2$ | 0.026 | 0.021 | 0.025 | 0.011 |

◂ Bonus Vs Non Bonus Days ◂ Purchases Vs Sales